

A data-efficient strategy for building high-performing medical foundation models

Received: 10 January 2024

Accepted: 4 February 2025

Published online: 5 March 2025



Yuqi Sun^{1,2}, Weimin Tan^{1,2}, Zhuoyao Gu¹, Ruian He¹, Siyuan Chen¹,
Miao Pang¹ & Bo Yan¹✉

Foundation models are pretrained on massive datasets. However, collecting medical datasets is expensive and time-consuming, and raises privacy concerns. Here we show that synthetic data generated via conditioning with disease labels can be leveraged for building high-performing medical foundation models. We pretrained a retinal foundation model, first with approximately one million synthetic retinal images with physiological structures and feature distribution consistent with real counterparts, and then with only 16.7% of the 904,170 real-world colour fundus photography images required in a recently reported retinal foundation model (RETFound). The data-efficient model performed as well or better than RETFound across nine public datasets and four diagnostic tasks; and for diabetic-retinopathy grading, it used only 40% of the expert-annotated training data used by RETFound. We also support the generalizability of the data-efficient strategy by building a classifier for the detection of tuberculosis on chest X-ray images. The text-conditioned generation of synthetic data may enhance the performance and generalization of medical foundation models.

Medical foundation models have recently emerged in several fields, such as ophthalmology¹, network biology², pathology image analysis³ and chest radiology⁴, providing a new direction for the development of medical artificial intelligence (AI) that can be applied to numerous downstream tasks. Medical foundation models are pretrained on massive amounts of data to extract generalizable feature representations. Researchers can easily adapt these models to specific tasks by fine-tuning them with small amounts of data with explicit labels⁵. Nevertheless, the demand for massive amounts of medical data poses substantial challenges to the development of medical foundation models^{5,6}. Acquiring massive amounts of medical data is extremely expensive and time-consuming, especially in fields requiring high-precision equipment, such as magnetic resonance imaging (MRI), and long-term patient tracking, such as oncology⁷ and neurodegenerative diseases⁸. In addition, large-scale pretraining medical data are typically collected from multiple centres into a centralized institution, which greatly increases the risk of exposing patient privacy because the centralized centre has access to a rich set of personal

patient information^{9–11}. Given these challenges, it is critical to develop a pretraining paradigm with high data efficiency for building medical foundation models from limited real-world pretraining datasets. This approach can effectively mitigate several issues that currently impede the development of medical foundation models, including data scarcity, extensive resource requirements and privacy concerns.

Generative AI aims to learn the underlying distribution of a given dataset using deep neural networks and generates new samples that resemble the original data¹². Various generative modelling architectures, such as variational autoencoders (VAEs)¹³, generative adversarial networks (GANs)¹⁴ and diffusion models^{15–17}, have been widely applied to medical fields for the discovery and design of new data, such as protein sequence generation^{18–20}, drug discovery^{21–23}, molecular design^{24–27} and medical imaging^{28–30}. Recent advances in generative AI, such as GLIDE³¹, DALL-E³² and stable diffusion¹⁶, have integrated natural language processing models with image processing models to generate realistic images corresponding to textual prompts³³. Compared with traditional generative models, these controllable multimodal models

¹Shanghai Key Laboratory of Intelligent Information Processing, School of Computer Science, Fudan University, Shanghai, China. ²These authors contributed equally: Yuqi Sun, Weimin Tan. ✉e-mail: byan@fudan.edu.cn

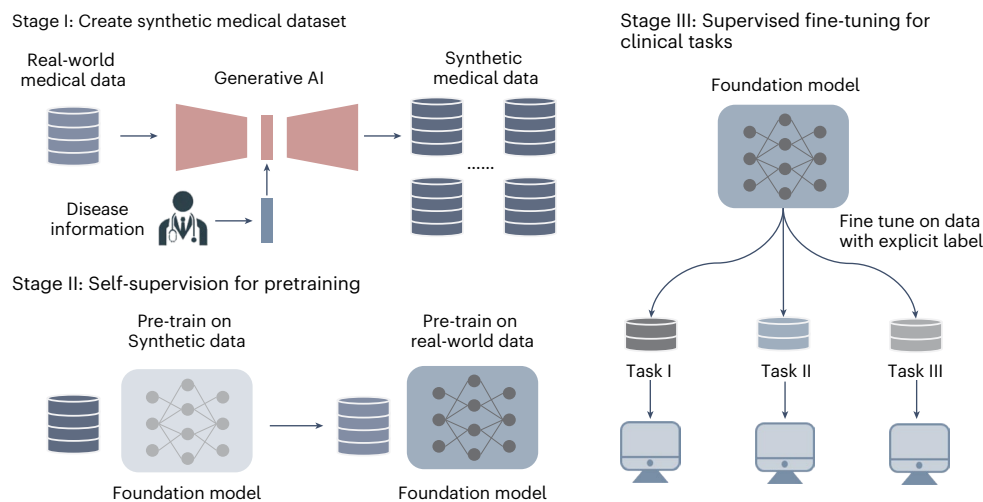


Fig. 1 | Overview of data-efficient strategy for building medical foundation models with generative AI. Stage one creates large-scale synthetic medical data from controllable generative AI trained on real-world medical data. Stage two builds the foundation model by two-phase pretraining on synthetic and real data

using self-supervised learning technique. Stage three adapts the foundation model to downstream tasks using supervised fine-tuning on data with explicit labels.

have been trained on millions of image–text pairs and possess flexible zero-shot generation capabilities and valuable medical information in their latent representations³⁴. By incorporating precise medical cues into these models, researchers can flexibly generate high-quality and diverse medical images in specific healthcare areas^{35–38}. In addition, the generated medical images offer notable advantages in protecting patient privacy since they do not directly reflect information about any real individual^{39,40}. These characteristics inspire us to create large-scale synthetic datasets using controllable generative AI, thereby alleviating the extensive data collection effort required for building medical foundation models.

In exploring methods to improve data efficiency in building medical foundation models using generative AI, ophthalmology AI is a notable option for the following reasons: eye care poses considerable challenges worldwide, as at least 2.2 billion people worldwide experience vision impairments according to the World Health Organization⁴¹; the latest retinal foundation model, RETFound¹, which is pretrained on millions of real-world retinal images, serves as a benchmark to validate the effectiveness of using generated data for building medical foundation models; retinal images illustrate clinical features of a wide range of ocular diseases (over 30 types⁴²), which can effectively validate the generalization ability of medical foundation models to different diseases; and several studies have demonstrated that generated retinal images are valuable for data argumentation in specific tasks, such as vessel segmentation⁴³ and age-related macular degeneration (AMD) grading³⁵.

In this work, we propose a data-efficient strategy that develops foundation models from a limited amount of medical data by leveraging controllable generative AI. The workflow is shown in Fig. 1, and consists of the following stages: in the first stage, we integrate domain-specific disease information into controllable generative AI by fine-tuning it on real-world medical data and then create a large-scale synthetic medical dataset using disease concepts as conditions; in the second stage, the foundation model is successively pretrained on the generated and real medical data using the self-supervised learning (SSL) technique to warm up the model parameters and learn accurate medical representations; in the final stage, the foundation model is adapted to specific tasks by supervised fine-tuning with explicit labels.

Following this strategy, we build a new retinal foundation model in ophthalmology called RETFound-DE, using only 16.7% (150,786 images, denoted as 150k) of the real-world colour fundus photography (CFP) images required in RETFound¹ (904,170 images, denoted as 900k).

Information on these images is provided in Supplementary Table 1. We use the latest multimodal generative AI, stable diffusion¹⁶, to learn the distribution relationship between CFP images and retinal disease concepts (such as ‘proliferative diabetic retinopathy’ and ‘branch retinal vein occlusion’) and create a synthetic dataset with ~1 million CFP images (Fig. 2a). After pretraining on large-scale synthetic and limited real-world retinal images, RETFound-DE achieves comparable or even superior performance to RETFound when adapted to nine public datasets across four downstream tasks, including diabetic retinopathy grading, glaucoma diagnosis, AMD grading and multidisease classification (Fig. 2b). We also validate the performance of RETFound-DE through external evaluation, label efficiency and fine-tuning time efficiency, demonstrating the excellent competitiveness of RETFound-DE as a medical foundation model in ophthalmology. In addition, we conduct variable-control experiments to illustrate the positive effect of synthetic retinal images on the performance and efficiency of developing RETFound-DE. Finally, we also conduct experiments on chest X-ray images to further demonstrate the generalizability of our data-efficient strategy.

Results

Model architectures and experimental setting

By using stable diffusion to model the joint distribution of retinal disease concepts and images, we flexibly generate retinal images with diverse clinical manifestations based on various medical concepts. Specifically, we use disease labels as text prompts to organize 150,786 retinal image–text pairs and fine tune Stable Diffusion v.1.4 to form retinal diffusion models. On the basis of the retinal diffusion model, we create a synthetic dataset with ~1 million CFP images using disease labels as conditions. The synthetic retinal image dataset comprises the 23 most common single-disease categories and 17 multidisease categories. The text prompts for real and synthetic retinal images and the corresponding number of images are provided in Supplementary Tables 2 and 3. We compare the performance of RETFound-DE against that of the following foundation models: SSL-ImageNet and RETFound. To obtain a fair comparison, the same network architectures (vision transformer⁴⁴), SSL pretraining technique (masked autoencoder⁴⁵) and fine-tuning process for downstream tasks are used in each model. SSL-ImageNet is pretrained on ImageNet-1k (~1.4 million natural images). RETFound initializes the model with the parameters of SSL-ImageNet and is further pretrained on 900k real CFP retinal images. RETFound-DE adopts

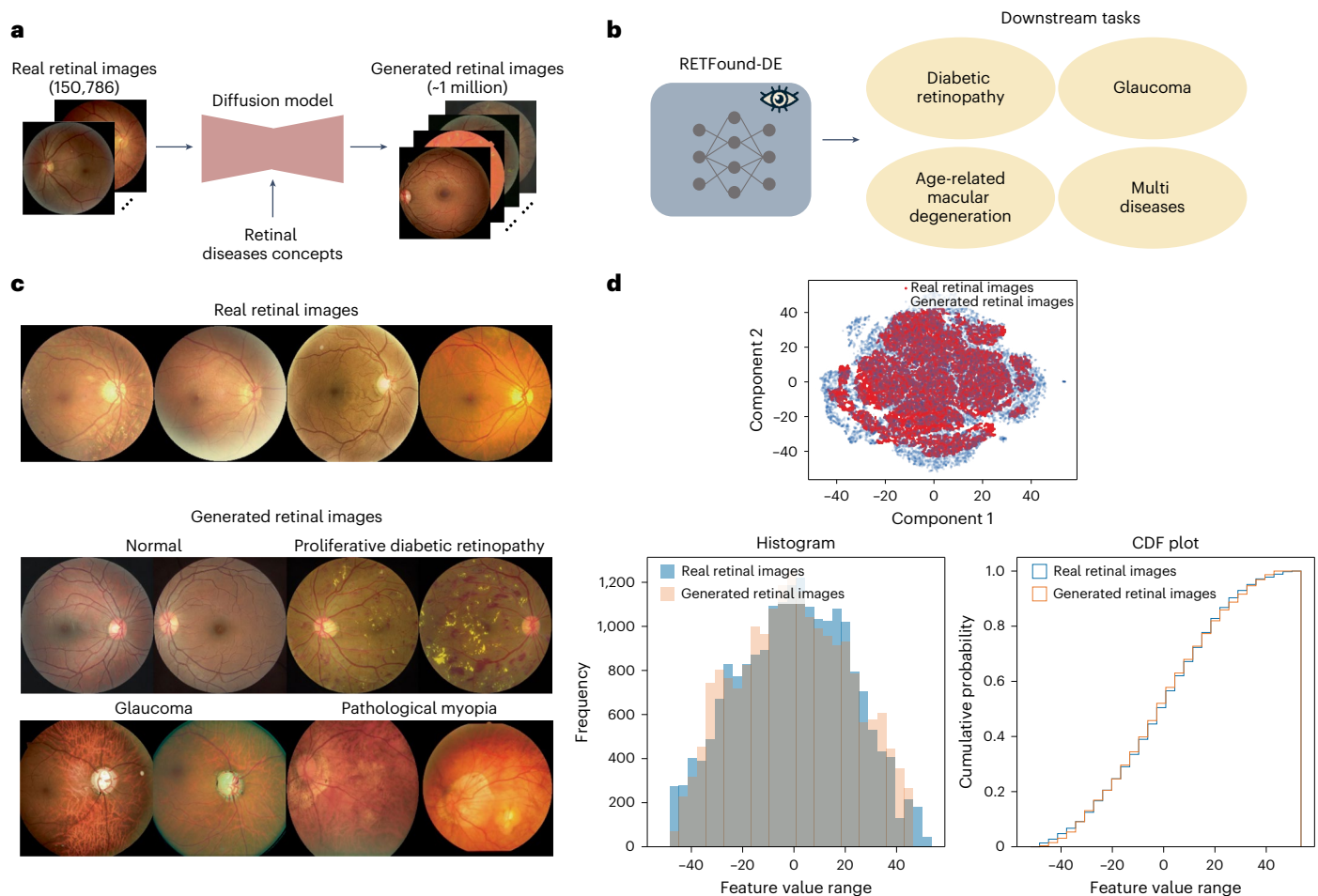


Fig. 2 | Schematic of RETFound-DE and consistency between real and generated retinal images in image and feature spaces. a, We used stable diffusion to model the conditional distribution of retinal images and disease concepts, and created a synthetic dataset with ~1 million CFP images.

b, Downstream tasks for RETFound-DE. We evaluated the performance of RETFound-DE in adapting to four ocular disease detection tasks, including diabetic retinopathy grading, glaucoma diagnosis, age-related macular degeneration grading and multidisease classification. **c**, Visual illustration of real and generated retinal images. We used diffusion to generate retinal

images with diverse clinical manifestations based on various disease labels. The generated retinal images possess the same physiological structures as real retinal images, such as blood vessels, optic discs and pathological features. **d**, *t*-SNE visualization, histograms and CDF plots of features from real and generated retinal images. We used RETFound to extract features. The results show that the real and generated retinal images have similar distribution in feature space. We performed 5 rounds of random sampling, each drawing 20,000 images from the real and synthetic datasets, then calculated the average distribution of these to prevent sampling bias.

the same initialization strategy as RETFound but is successively pretrained on ~1 million generated retinal images and 150k real retinal images. We evaluate the performance of RETFound-DE in adapting to nine publicly available datasets across four ocular disease detection tasks, including (1) diabetic retinopathy grading, (2) glaucoma diagnosis, (3) AMD grading and (4) multidisease classification. The information on downstream datasets is reported in Supplementary Table 4. The area under the receiver operating curve (AUROC) and the area under the precision-recall curve (AUPR) are obtained for evaluating the performance of the differing models. For each downstream task experiment, we train the model with five different seeds and evaluate its performance to obtain five results for statistical analysis. We calculate the *P* values between RETFound-DE and RETFound using two-sided *t*-tests for each task to determine the presence of statistically significant differences ($P < 0.05$ indicates a significant difference).

Consistency between real and generated retinal images

The generated retinal images possess the same physiological structures as real retinal images, such as blood vessels and optic discs, and pathological features such as hard exudates in diabetic retinopathy (Fig. 2c). To measure the consistency of feature distribution between

generated and real retinal images, we randomly sampled 20,000 real and 20,000 generated retinal images, extracted features using RETFound and applied *t*-distributed stochastic neighbor embedding (*t*-SNE), a dimensionality reduction technique, to visualize and compare the feature distributions in a lower-dimensional space. We performed five rounds of random sampling and calculated the average distribution of these samples to prevent sampling bias. Figure 2d visualizes the differences in feature distribution using *t*-SNE visualization, histograms and cumulative distribution function (CDF) plots, which show no substantial difference in the feature distribution between the generated and real retinal images. The consistency in pathological structures and feature distributions provides a sound basis for employing generated retinal images in the subsequent self-supervised pretraining stage.

Performance in downstream tasks

We evaluated the performance of RETFound-DE across four ocular disease detection tasks. The AUROC result is reported in Fig. 3a and the AUPR result is reported in Extended Data Fig. 1a. We also show the confusion matrices of several datasets in Extended Data Fig. 2. All quantitative results are listed in Supplementary Tables 5 and 6.

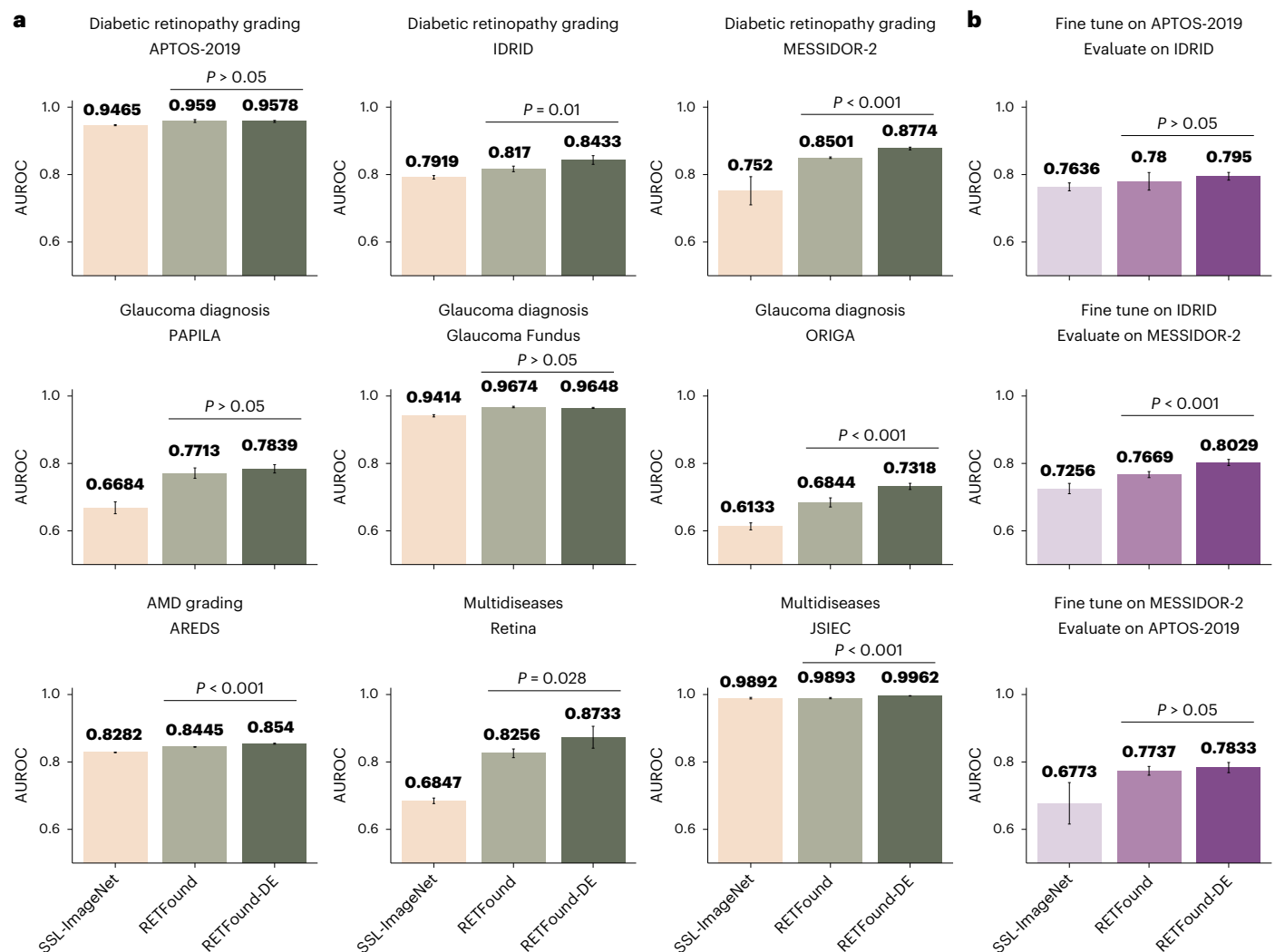


Fig. 3 | Performance in downstream tasks. a, Internal evaluation. We fine-tuned the pretrained models on 9 public datasets across 4 downstream tasks: diabetic retinopathy grading, glaucoma diagnosis, AMD grading and multidisease classification. Compared with RETFound, RETFound-DE achieved significantly superior performance on 6 datasets ($P < 0.05$) and comparable performance on the other 3 datasets ($P > 0.05$). **b, External evaluation.** The foundation models

were fine-tuned on one diabetic retinopathy grading dataset and evaluated on the other dataset. RETFound-DE significantly outperformed RETFound when fine-tuned on IDRID and evaluated on MESSIDOR-2 ($P < 0.001$). We present the mean $\pm$ 95% CI of AUROC. The listed P values were calculated using two-sided t -test.

Diabetic retinopathy (DR) is a widespread eye disease that affects people with diabetes. Diabetic retinopathy grading aims to detect DR at earlier stages, allowing for timely interventions to prevent progression and preserve vision. We adapted RETFound-DE to three diabetic retinopathy grading tasks: Kaggle APTOS-2019 (ref. 46) ($n = 3,662$), IDRID⁴⁷ ($n = 516$) and MESSIDOR-2 (ref. 48) ($n = 1,744$) datasets. Each image in these datasets is labelled on the basis of the 5-stage International Clinical Diabetic Retinopathy Severity scale (from no diabetic retinopathy to proliferative diabetic retinopathy). Compared to RETFound, RETFound-DE achieves comparable performance on the Kaggle APTOS-2019 dataset with an AUROC of 0.9578 (95% confidence interval (CI) 0.9549, 0.9608). On IDRID and MESSIDOR-2 datasets, RETFound-DE significantly outperforms RETFound with an AUROC of 0.8433 (95% CI 0.8304, 0.8561) and 0.8774 (95% CI 0.8734, 0.8814), respectively, especially on the MESSIDOR-2 dataset ($P < 0.001$). The AUPR results of RETFound-DE are also competitive with RETFound on these three datasets (Extended Data Fig. 1a).

Glaucoma is a heterogeneous group of degenerative neural disorders characterized by elevated intraocular pressure damage to the optic nerve and loss of peripheral vision. It is the leading cause of

irreversible blindness worldwide. We adapted RETFound-DE to three glaucoma diagnosis datasets: PAPILA⁴⁹ ($n = 489$), Glaucoma Fundus ($n = 1,544$) and ORIGA⁵⁰ ($n = 650$), which are labelled as no glaucoma, glaucoma or suspected glaucoma. RETFound-DE achieves comparable performance on the PAPILA and Glaucoma Fundus datasets and significantly superior performance on the ORIGA dataset ($P < 0.001$), with an AUROC of 0.7319 (95% CI 0.7228, 0.741). The AUPR results on the PAPILA and ORIGA datasets also show that RETFound-DE significantly outperforms RETFound ($P = 0.035$ and $P < 0.001$, Extended Data Fig. 1a).

Age-related macular degeneration is an eye disorder arising from degeneration of the macula, which is the central region of the retina responsible for precise and forward-focused vision. Primarily associated with aging, AMD has emerged as a leading contributor to visual impairment among the elderly population. We performed AMD grading by fine-tuning RETFound-DE in the Age-Related Eye Disease Study (AREDS), a long-term multicentre and prospective study of the clinical course of AMD. AREDS proposed a simplified AMD clinical classification system with four categories: non-AMD, mild, moderate and advanced AMD. To validate the performance of RETFound-DE on a limited annotated dataset, we randomly selected 800 patients,

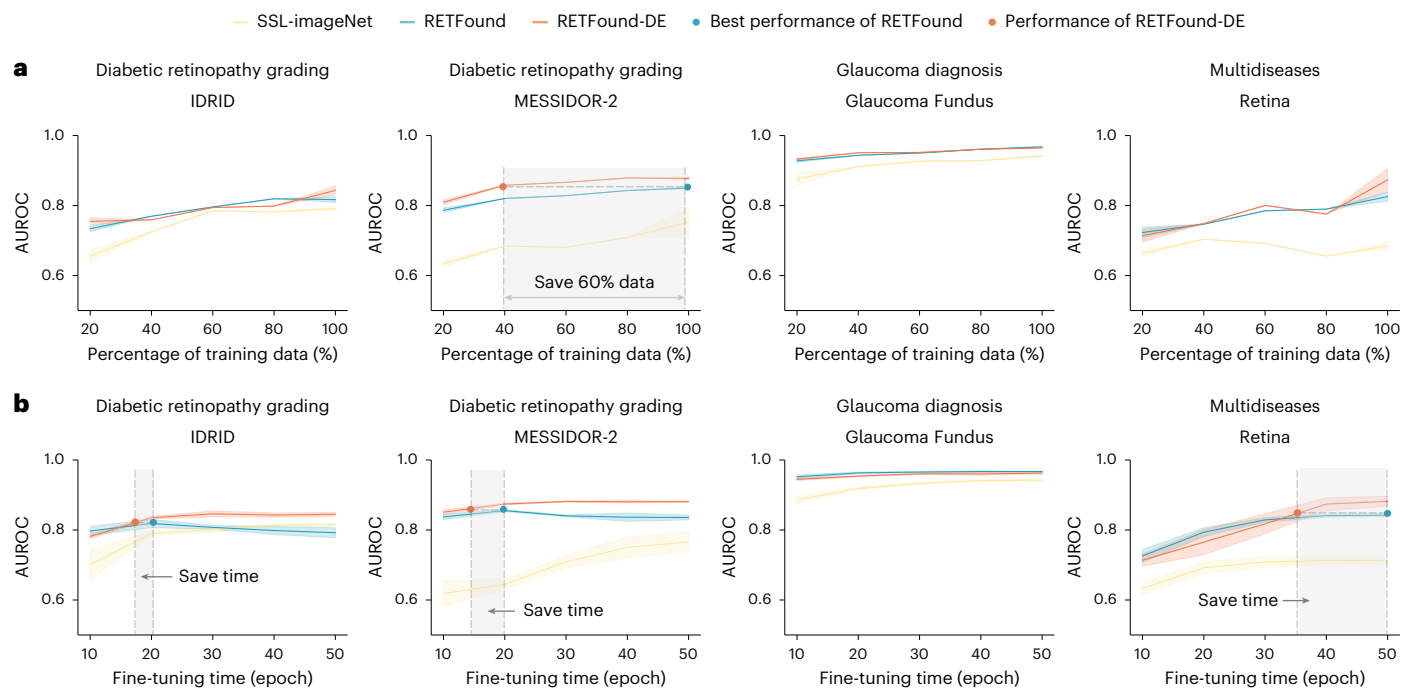


Fig. 4 | Labelling and fine-tuning time efficiency. **a**, Labelling efficiency refers to the amount of training data and labels needed to achieve the target performance for foundation models when adapted to downstream tasks. RETFound-DE and RETFound show comparable labelling efficiencies on IDRID, Glaucoma Fundus and Retina datasets. On the MESSIDOR-2 dataset, RETFound-DE achieves better performance than RETFound, with a saving of 60% labelled training data. **b**, Fine-tuning time efficiency refers to the fine-tuning time needed to achieve

the target performance for foundation models when adapted to downstream tasks. RETFound-DE and RETFound show comparable fine-tuning time efficiency on the Glaucoma Fundus dataset. On IDRID, MESSIDOR-2 and Retina datasets, RETFound-DE achieves better performance than RETFound in less time. Coloured lines indicate the mean value of AUROC and the 95% CIs are visualized as coloured bands. Dashed grey lines highlight the difference between two models.

comprising a total of 5,768 images. The AUROC and AUPR results show that RETFound-DE achieves significantly better performance ($P < 0.001$).

In addition to specific ocular diseases, recent deep-learning research⁴² has focused on detecting multiple ocular diseases. We evaluated the performance of RETFound-DE in multidisease classification on the Retina ($n = 601$) and JSIEC ($n = 1,000$)⁴² datasets. The Retina dataset contains four categories: normal, cataract, glaucoma and other diseases. RETFound-DE achieves significantly better AUROC and AUPR results (0.8733 and 0.7478) than RETFound (0.8256 and 0.6385). The confusion matrix of the Retina dataset (Extended Data Fig. 2) shows that RETFound-DE achieves higher detection accuracy in each category. The JSIEC dataset encompasses 39 different disease types, such as diabetic retinopathy, glaucoma, AMD, cataract, retinal vein occlusion and retinal artery occlusion. RETFound-DE achieves an AUROC of 0.9962 (95% CI 0.9957, 0.9968) and an AUPR of 0.9074 (95% CI 0.8993, 0.9155), which also significantly outperforms RETFound ($P < 0.001$).

Labelling and fine-tuning time efficiency

We then compared RETFound-DE with SSL-ImageNet and RETFound in terms of labelling efficiency and fine-tuning time efficiency, which refer to the explicitly labelled training data and training time required for a deep-learning model to achieve the target performance. High labelling efficiency and fine-tuning time efficiency can reduce the annotation workload of experts and the computational costs during the fine-tuning process. Figure 4a shows that RETFound-DE achieves comparable labelling efficiency to RETFound on IDRID, Glaucoma Fundus and Retina datasets. Notably, on the MESSIDOR-2 dataset, RETFound-DE achieves better performance than RETFound, with a saving of 60% of the annotated training data. The AUPR results (Extended Data Fig. 3a) also show that RETFound-DE achieves comparable labelling efficiency to RETFound. We show the fine-tuning time efficiency in Fig. 4b. RETFound-DE shows comparable or even higher fine-tuning

time efficiency than RETFound. They are comparable on the Glaucoma Fundus dataset, and RETFound-DE achieves better performance than RETFound in less time on IDRID, MESSIDOR-2 and Retina datasets. The AUPR results of fine-tuning time efficiency are shown in Extended Data Fig. 3b. The dashed grey lines highlight the difference between RETFound and RETFound-DE.

External evaluation in downstream tasks

The generalizability of medical AI is crucial for its application in clinical settings, as it involves handling ‘out-of-distribution’ data that are not seen during training, such as data from different populations and devices. We evaluated the generalization ability of the foundation model through external evaluation, which means fine-tuning the model on one dataset and then directly evaluating it on another dataset from a different data centre. We first compared RETFound-DE with SSL-ImageNet and RETFound in the diabetic retinopathy grading task, as they were both labelled on the basis of the 5-stage International Clinical Diabetic Retinopathy Severity scale. Figure 3b shows that RETFound-DE achieved comparable generalization ability to RETFound in two external experiments ($P > 0.05$). When fine-tuned on IDRID and evaluated on the MESSIDOR-2 dataset, RETFound-DE achieved an AUROC of 0.8029 (95% CI 0.7939, 0.8119), significantly higher than 0.7669 (95% CI 0.758, 0.7758) for RETFound ($P < 0.001$). The AUPR result of the external evaluation also demonstrates the excellent generalization capabilities of RETFound-DE (Extended Data Fig. 1b).

To demonstrate the value of using large-scale synthetic data on generalizability, we also compared RETFound-DE (denoted as ‘With synthetic data’) with a baseline model, SSL-ImageNet-Retinal (150k) (denoted as ‘Without synthetic data’), which was only pretrained on 150k real retinal images. Figure 5 shows the external validation of ‘With synthetic data’ and ‘Without synthetic data’. We conducted external validation on 8 datasets from 6 different data centres

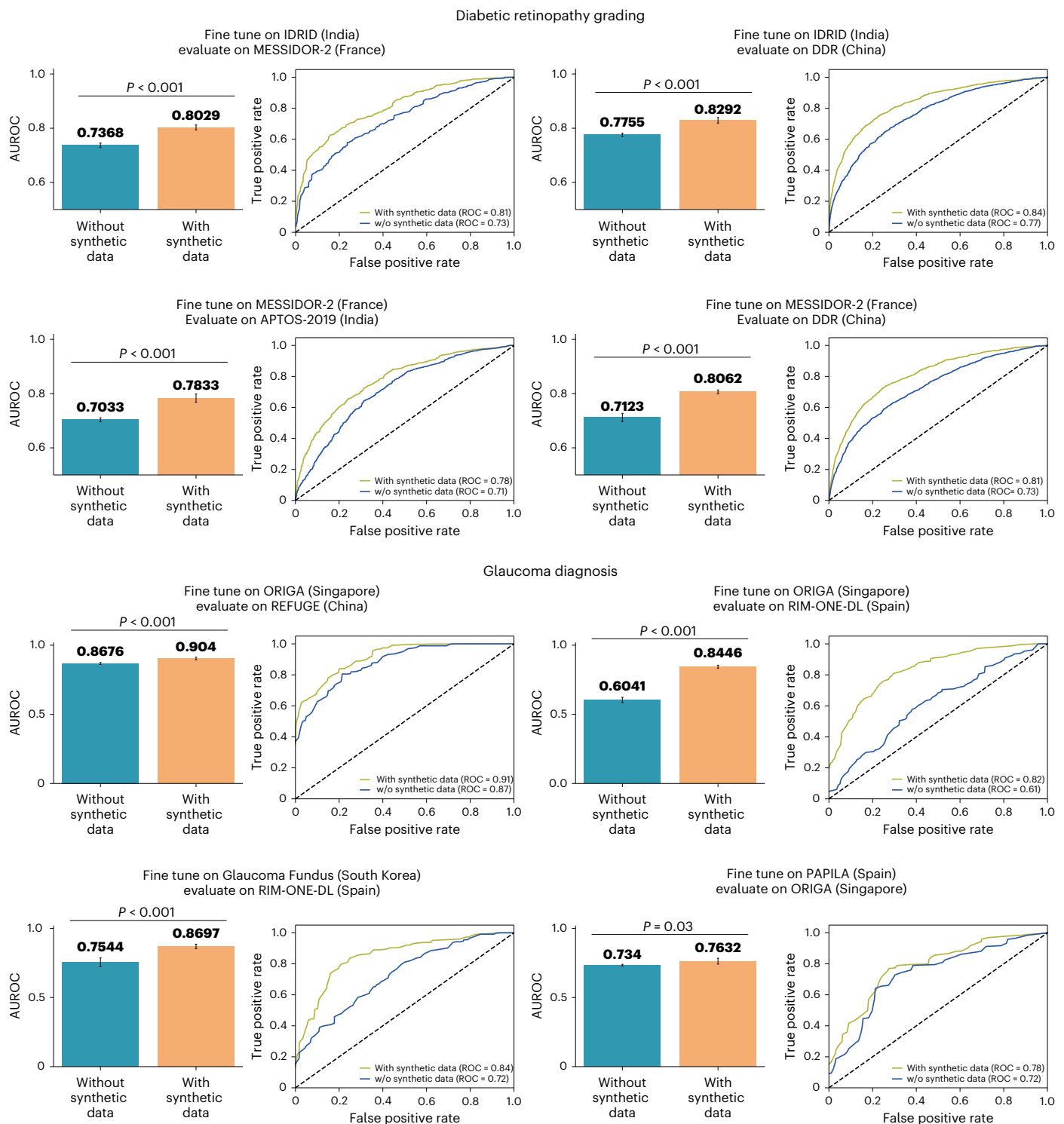


Fig. 5 | Cross-centre external evaluation for models pretrained with and without synthetic data. ‘With synthetic data’ denotes our foundation model RETFound-DE, which was pretrained on large-scale synthetic data and 150k real data. ‘Without synthetic data’ denotes our baseline model SSL-ImageNet-Retinal (150k) that was only pretrained on 150k real retinal images. We conducted external validation on 8 datasets from 6 different data centres worldwide, involving diabetic retinopathy grading and glaucoma diagnosis tasks. The pretrained model was fine-tuned on dataset from one country and evaluated

directly on a dataset from another country. The results show that RETFound-DE (with synthetic data) significantly outperforms SSL-ImageNet-Retinal (150k) in all cross-centre external validations ($P < 0.001$ on 7 tasks and $P = 0.03$ on 1 task). Left: we report the statistical results of two pretrained models fine-tuned with 5 different random seeds in the bar charts and present the mean $\pm$ 95% CI of AUROC. P values were calculated using two-sided t -tests. Right: we also report results in ROC curves for two models trained using one of the five random seeds.

worldwide, involving diabetic retinopathy grading and glaucoma diagnosis tasks. The results show that ‘With synthetic data’ significantly outperforms ‘Without synthetic data’ in all cross-centre external validations

($P < 0.001$ on 7 tasks and $P = 0.03$ on 1 task). These experiments demonstrate that using large-scale synthetic data for pretraining can enhance the robustness and generalizability of foundational models.

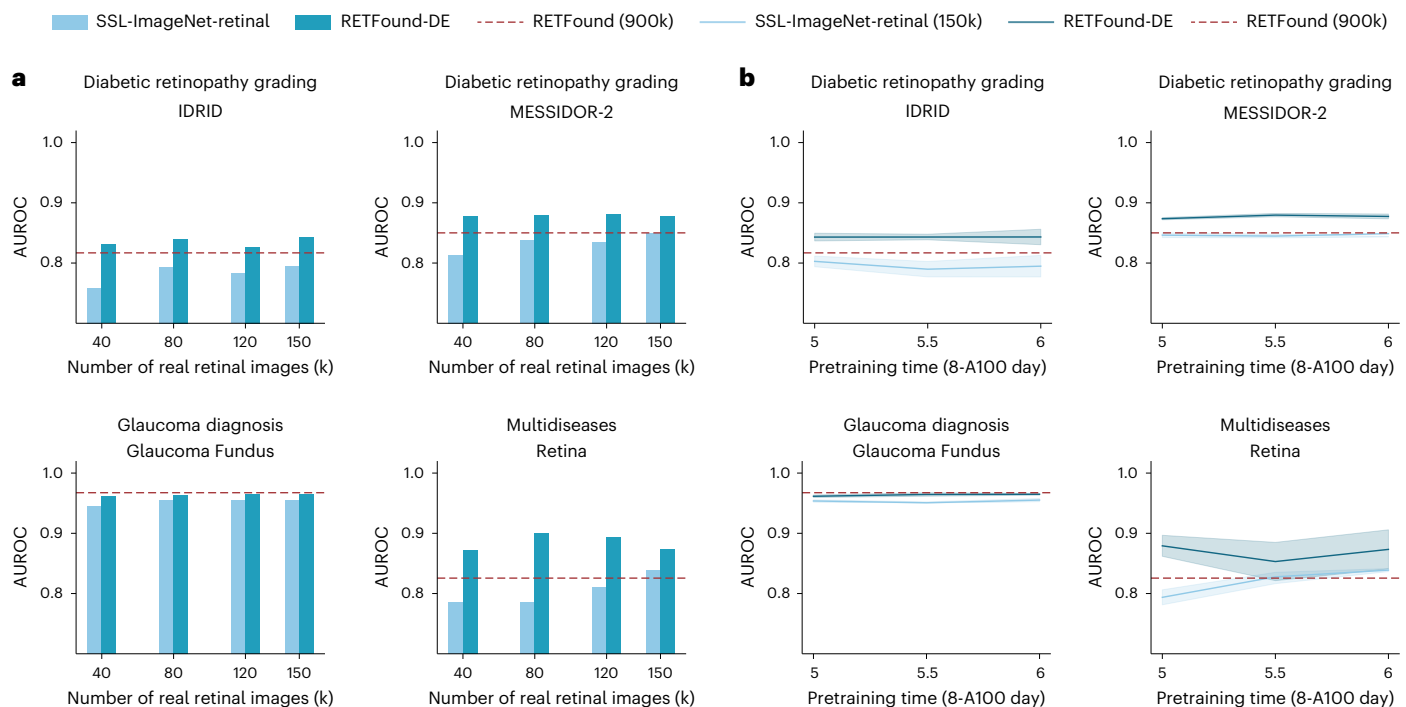


Fig. 6 | Data and pretraining time efficiency. a, Effect of the synthetic data on using different amounts of real data for pretraining. The performance of SSL-ImageNet-Retinal progressively increases on four downstream tasks as the quantity of real retinal images used for pretraining increases. By pretraining on generated images, RETFound-DE shows a notable performance improvement compared with SSL-ImageNet-Retinal. On IDRID, MESSIDOR-2 and Retina datasets, RETFound-DE outperforms RETFound when pretrained on only 40k real retinal images. **b**, Performance of RETFound-DE and SSL-ImageNet-Retinal

(150k) over a matched computational time from five to six 8-A100 days. We use 8-A100 day as a unit to denote the pretraining time of using 8 NVIDIA A100 GPUs for 1 day. For both models, the pretraining dataset is the 150k real retinal image dataset at this training phase. RETFound-DE consistently outperforms SSL-ImageNet-Retinal (150k) on all four downstream datasets within the same pretraining time. Red dashed lines indicate the AUROC performance of RETFound. Coloured lines indicate the mean value of AUROC and the 95% CIs are visualized as coloured bands.

Pretraining data and time efficiency

We further validated the necessity of using synthetic data for pretraining in terms of data and pretraining time efficiency, which refers to the number of real retinal images and pretraining time required for building the foundation model. We compared RETFound-DE with the baseline model SSL-ImageNet-Retinal, which follows the same pretraining paradigm as RETFound but on limited real retinal images (from 40k to 150k). Figure 6a shows the performance comparison between RETFound-DE and SSL-ImageNet-Retinal in downstream tasks after pretraining on different numbers of real retinal images. The performance of SSL-ImageNet-Retinal progressively increases on four downstream tasks as the quantity of real retinal images used for pretraining increases. Compared with SSL-ImageNet-Retinal, RETFound-DE shows a notable performance improvement in each task. In particular, RETFound-DE outperforms RETFound on IDRID and MESSIDOR-2 datasets when pretrained on only 40k real retinal images, indicating that pretraining on synthetic data can enhance the medical representation of the foundation model on a limited number of data. The AUPR results are shown in Extended Data Fig. 4a.

We also compared the performance of SSL-ImageNet-Retinal (150k) and RETFound-DE in downstream tasks after different pretraining times. We use 8-A100 day as a unit to represent the pretraining time of using 8 NVIDIA A100 GPUs for 1 day. RETFound-DE was pretrained on synthetic images (~1 million) for five 8-A100 days and then on 150k real retinal images for one 8-A100 day (six 8-A100 days in total). For a fair comparison, we pretrained SSL-ImageNet-Retinal (150k) on 150k real images for six 8-A100 days. Figure 6b shows the performance of SSL-ImageNet-Retinal (150k) and RETFound-DE over a matched pretraining time of five to six 8-A100 days. For both models, the pretraining dataset is the 150k real retinal image dataset at this period.

RETFound-DE consistently outperforms SSL-ImageNet-Retinal (150k) at different pretraining times. The AUPR results are shown in Extended Data Fig. 4b. We also show the pretraining loss of RETFound-DE and SSL-ImageNet-Retinal (150k) on synthetic and 150k real retinal images (Supplementary Fig. 1). After being pretrained on synthetic data, RETFound-DE was pretrained on 150k real retinal images for 200 epochs (one 8-A100 day) and converged to the same loss as SSL-ImageNet-Retinal (150k), which was pretrained for 1,200 epochs (six 8-A100 days).

Exploring the usage of synthetic data

In the following, we discuss the specifics of using synthetic data to provide a comprehensive perspective on the usage of generated data and help researchers better understand and assess the pros and cons of using generated data in building medical foundation models. The specific experimental setup and results are presented in Supplementary Notes. We first analysed the effects of a balanced distribution of synthetic data on the performance of medical AI models. The results show that the balanced synthetic dataset helps improve model performance on underrepresented diseases in the real unbalanced dataset (Supplementary Note 1 and Figs. 2 and 3). We then explored the relationship between the number of generated images and model performance. We found that the performance of the pre-trained model gradually improved with increase in the size of synthetic data, but when the data size reached 900k and 1,200k, the performance and training loss showed little changes (Supplementary Note 2 and Figs. 4a and 5). In addition, we investigated the lowest number of real images required to maintain robust model performance and found that the pretrained model exhibited poor performance with 100 and 1,000 images, and highly improved when the number reached 10,000

(Supplementary Note 3 and Fig. 4b). Finally, we analysed the effect of using different versions of the stable diffusion (SD) model on the quality of generated images (Supplementary Note 4, Table 7 and Fig. 6). We show that various versions of the stable diffusion model, including SDv.1.4, SDv.2.0 and SDXL, can generate realistic retinal images.

Experiments on chest X-ray images

We conducted experiments on chest X-ray (CXR) images to explore the potential of our data-efficient strategy in extending to other medical domains. Please refer to Supplementary Note 5 for more details. We first selected 100,000 images from the CheXpert⁵¹ dataset and applied self-supervised learning (SSL) to pretrain a foundational model, named CXRFM, as our benchmark. Subsequently, we selected 20,000 images (20% of images used in CXRFM) to train a CXR generative model, and generated 80,000 synthetic CXR images. Following our strategy, we sequentially pretrained on synthetic and real data to build a model, which we named CXRFM-DE. To assess the performance and generalization of CXRFM-DE, we performed internal and external validation on two tuberculosis-related datasets: ShenzhenCXR⁵² and TBChest⁵³. Our internal validation showed that CXRFM and CXRFM-DE performed similarly on ShenzhenCXR and significantly outperformed SSL-ImageNet, while all models achieved high performance on TBChest (Extended Data Fig. 5a). External validation revealed that when fine-tuned on TBChest and evaluated on ShenzhenCXR, CXRFM and CXRFM-DE performed comparably, both surpassing SSL-ImageNet. When fine-tuned on ShenzhenCXR and evaluated on TBChest, CXRFM outperformed CXRFM-DE (Extended Data Fig. 5b). Finally, we examined the impact of synthetic data on model generalization by comparing CXRFM-DE (denoted as ‘With synthetic data’) to SSL-ImageNet-CXR (20k), which was pretrained only on 20,000 real CXR images (denoted as ‘Without synthetic data’). CXRFM-DE significantly outperformed SSL-ImageNet-CXR (20k), demonstrating the value of synthetic data in enhancing model generalization (Extended Data Fig. 5c). These findings align with our previous results on retinal images, underscoring the potential for generalization of our strategy to other medical fields.

Discussion

In this study, we introduce a data-efficient strategy that uses controllable generative AI to create large-scale synthetic datasets, providing an effective solution to alleviate the extensive data collection efforts required to build medical foundation models. We collected 150,786 real CFP retinal images (only 16.7% of images in RETFound) and created a synthetic dataset with ~1 million CFP images. After successively pretraining on synthetic and real datasets, we built a new retinal foundation model in ophthalmology, RETFound-DE. Our experimental evaluation provides extensive evidence that RETFound-DE shows comparable or superior capabilities to RETFound in enhancing model performance, generalizability, labelling and fine-tuning time efficiency when adapted to 9 datasets across 4 downstream tasks, including diabetic retinopathy grading, glaucoma diagnosis, AMD grading and multidisease classification. We also provide strong evidence for model performance benefits from pretraining on large-scale synthetic data in terms of performance, generalizability, pretraining data and time efficiency. Our data-efficient strategy provides valuable insights for many medical fields that were previously unable to build foundation models due to limited data.

Our work has been encouraged by the success of many generative AI applications in medical healthcare.

Many studies^{35,43,54,55} leverage generative AI to synthesize realistic images for data augmentation to enhance supervised model performance in specific tasks. Most recently, multimodal generative AI has been adopted in auditing the inference processes²⁹ and improving fairness⁵⁶ of medical classifiers. These efforts demonstrate the potential of generative AI in building interpretable, high-performance and fairer medical AI models. Our work integrates generative AI into the construction process of medical foundation models by creating a large-scale

synthetic dataset for self-supervised pretraining. The synthetic dataset comprises 912,088 images (72%) across 23 single-disease categories and 352,907 images (28%) across 17 multidisease categories, totalling 1,264,995 images. The single-disease category includes the 23 most common diseases in the real retinal image dataset, such as referable glaucoma, diabetic retinopathy, age-related macular degeneration and branch retinal vein occlusion. In addition, we synthesize images with multiple diseases by creating textual combinations of single-disease categories, such as ‘referable glaucoma, mild non-proliferative diabetic retinopathy’. We also perform automatic image quality assessment by comparing the feature consistency between the generated and real images, demonstrating the feasibility of using generated images for pretraining (Fig. 2d). The synthetic dataset includes diverse, realistic and canonical images, which provides great support for improving the performance (Fig. 3) and generalization (Fig. 5) of the foundation model.

In the pretraining phase, RETFound-DE employs MAE^{45,57} (masked autoencoder) to learn retina-specific representations from unlabelled retinal images. MAE aims to reconstruct the original complete retinal images from highly masked image patches with limited visible information, requiring the model to capture basic retinal structures (Supplementary Fig. 7a), such as blood vessels and optic nerves that are critical to the diagnosis of ocular diseases. RETFound initializes models with weights pretrained on ImageNet-1k (~1.4 million natural images) to enhance its representation of retinal images. RETFound-DE benefits from a more effective parameter initialization strategy through pretraining on an extensive dataset of generated retinal images with the same image structure as real retinal images. In comparison with SSL-ImageNet-Retinal, which is similar to RETFound but pretrained on a much smaller amount of data (150k), RETFound-DE achieves a substantial improvement in model performance (Fig. 6), generalization (Fig. 5) and convergence rate (Supplementary Fig. 1) when pretrained on real retinal images. In addition, our strategy does not require more pretraining time than the 800 epochs for RETFound. We pretrained RETFound-DE for 600 epochs (five 8-A100 days) on generated retinal images and 200 epochs (one 8-A100 day) on 150k real retinal images. More details of the training process are described in Methods.

Although only 150k real retinal images were involved in the pretraining process, RETFound-DE has demonstrated excellent performance as a foundation model in various medical applications. On nine datasets across four downstream retinal disease diagnosis tasks, RETFound-DE achieves comparable or superior performance to RETFound (Fig. 3a and Extended Data Fig. 1a). In particular, RETFound-DE significantly outperforms RETFound on IDR1D, MESSIDOR-2, ORIGA, AREDS, Retina and JSIEC datasets, demonstrating its powerful medical representation of retinal images. RETFound-DE improves the performance of ocular disease diagnosis by recognizing disease-related lesions in retinal images. For instance, we show that RETFound-DE can identify hard exudates in diabetic retinopathy and optic cups in glaucoma during the inference process (Supplementary Fig. 7b). Another important capability of the foundation model is the reduction of expert-annotation data and model training time required for downstream tasks. Figure 4 demonstrates the high labelling efficiency and fine-tuning time efficiency of RETFound-DE. On the MESSIDOR-2 datasets, it achieves the performance of RETFound using only 40% of the labelled training data. Our experiments on chest X-ray images further demonstrate the potential of our data-efficient strategy in extending to other medical domains.

Even though our results have demonstrated that generated data can effectively build high-performing medical AI models with limited real medical data, we suggest a need for prudence and careful consideration in using generated data. The recent findings⁵⁸ on AI model collapse when trained on recursively generated data highlight the potential dangers of over-reliance on synthetic content. While this study primarily focuses on the unconditional generation of text and

handwritten digital images, it raises concerns about the risks and challenges associated with generated data in more complex environments. We extend this discussion to the medical field, focusing on disease-label conditioned medical image generation. We discuss several potential risks and challenges below, which may provide insights for future researchers using large datasets of synthetic data for building medical AI models.

A primary challenge is the development of automated quality assessment methods for large-scale synthetic data. The quality of synthetic data is crucial for the reliability of medical AI models, as poor-quality data can lead to incorrect or irrelevant learning, negatively impacting clinical performance. Traditional methods that involve expert evaluation are impractical for large-scale synthetic data, as evaluating a small sample does not adequately represent the overall data quality, and evaluating large volumes of data consumes substantial time and resources. Therefore, it is essential to develop automated quality assessment methods for large-scale synthetic data. This involves establishing standardized quality metrics to ensure that generative models produce consistent and realistic images, which is vital for enhancing the credibility of synthetic data in medical AI applications.

Another concern is the potential bias introduced by using a large-scale synthetic dataset for pretraining. We analysed the bias issues in generated retinal images, using population distribution as an example, and found that models pretrained on synthetic data exhibit the same population biases as the real data used to generate these models (Supplementary Note 6 and Table 8). Furthermore, we explored methods to create bias-aware synthetic datasets using controlled generative AI by specifying sensitive attributes in images (Supplementary Note 7). For example, we incorporated age and gender information in the text prompts, such as 'Age 69, Sex Female, Disease: moderate non-proliferative retinopathy', which reflects both the disease condition and sensitive attributes. Extended Data Fig. 6 demonstrates the consistency of feature distribution between real and synthetic bias-aware datasets with respect to the different subgroups of age and gender. This bias-aware synthetic data generation method has been demonstrated in a recent study⁵⁶ to enhance the fairness of models addressing underrepresented diseases and minority groups. It is valuable since a recent report⁵⁹ found that the widespread bias in real medical datasets may lead to AI models frequently failing to maintain fairness with new, unseen data due to reliance on demographic shortcuts. We advise researchers to carefully consider the potential biases introduced by synthetic data. This involves developing generative models specifically designed to generate fair data, collecting more balanced real medical data and considering specific sensitive attributes that may influence underrepresented diseases and minority groups when creating synthetic datasets.

In addition, the potential for misuse and fabrication of synthetic data is an important concern in medical AI. Synthetic data could be used to fabricate research results or create fraudulent medical records, undermining trust in medical research and clinical decision-making. In addition, there is a risk that synthetic data might simulate unrealistic clinical scenarios, leading to inappropriate or harmful medical interventions. Ensuring the quality and relevance of synthetic data involves rigorous validation processes, regulatory oversight and education on ethical use. By addressing these challenges, we can harness the benefits of synthetic data while safeguarding the integrity of medical research and practice.

Finally, excessive dependence on synthetic data can lead to models that overfit synthetic scenarios and fail to generalize well to real-world data. This over-reliance can result in neglecting the collection and use of high-quality real-world data, which is crucial for validating and testing models in practical applications. In addition, synthetic data cannot create representations for groups or diseases absent in the real data. Therefore, it cannot expand the dataset to include these missing populations or conditions. This limitation highlights the critical need

for bolstering data sharing and collaboration between healthcare institutions and researchers to propel the advancement of general artificial intelligence models.

In summary, we have described a data-efficient strategy for using generative AI to scale up the pretraining dataset for building a retinal foundational model, RETFound-DE, from a limited collection of real retinal images (150k). When adapted for various ocular disease detection tasks, RETFound-DE showed great performance in the detection of eye disease, and in labelling and fine-tuning efficiency. Variable-controlling experiments demonstrate the positive impact of generated data on building RETFound-DE with limited pretraining datasets. Our work shows the potential for substantial cost savings in dataset acquisition for foundation-model development, and thus provides an effective solution for other diseases that were once discouraged from building foundation models owing to the limited data.

Methods

Datasets

We collected 150,786 CFPs from multiple publicly available datasets, including AIROGS⁶⁰, Kaggle EyePACS, DDR and ODIR-2019. Of these datasets, 101,442 images are from AIROGS, a large dataset with 54,274 patients for glaucoma screening. The age distribution of AIROGS has a mean value of 56.7 and standard deviation of 10.2. Totals of 35,126 and 7,218 images were obtained from Kaggle EyePACS and DDR, respectively, for diabetic retinopathy grading. A total of 7,000 images were from ODIR-2019, a multidisease classification dataset including cataract, myopia, diabetic retinopathy, AMD and glaucoma. Images in AIROGS and Kaggle EyePACS datasets are from the United States, while DDR and ODIR-2019 are from China. Various imaging devices are involved in these four datasets, including Canon CR1/DGi/CR2 (Canon), Optovue iCam (Optovue), Topcon NW (Topcon), DRS (Centervue), ZEISS and Kowa. We list the detailed information on real datasets in Supplementary Table 1. We adhere to all relevant guidelines and ethics regulations in our use of data.

Synthetic retinal images dataset

The synthetic retinal images dataset primarily comprised 912,088 images (72%) across 23 single-disease categories and 352,907 images (28%) across 17 multidisease categories, totalling 1,264,995 images. The single-disease category included the 23 most common diseases in the real retinal image dataset, including referable glaucoma, diabetic retinopathy, age-related macular degeneration, branch retinal vein occlusion and so on. In addition, some images in real datasets had multiple disease labels. We synthesized images with multiple diseases by creating textual combinations of single-disease categories, such as 'referable glaucoma, mild non-proliferative diabetic retinopathy'. We list the disease categories and image quantity for each category of the real synthetic dataset in Supplementary Tables 2 and 3.

Datasets for downstream tasks

We evaluated model performance on nine public datasets across four ocular disease detection tasks. Kaggle APTOS-2019 (India), IDRID (India) and MESSIDOR-2 (France) were used for diabetic retinopathy grading. Each image in these datasets was labelled on the basis of the 5-stage International Clinical Diabetic Retinopathy Severity scale, from no diabetic retinopathy to proliferative diabetic retinopathy. For glaucoma, PAPILA (Spain), Glaucoma Fundus (South Korea) and ORIGA (Singapore) were included. ORIGA labels each image with glaucoma and non-glaucoma, while PAPILA and Glaucoma Fundus has three category labels: non-glaucoma, early glaucoma (suspected glaucoma) and advanced glaucoma. For age-related macular degeneration, the AREDS was used in this study. AREDS includes over 134,500 CFPs from 4,757 participants and 4 AMD categories for each image: non-AMD, mild, moderate and advanced AMD. To evaluate the performance of RETFound-DE on limited labelled data, we randomly selected

5,768 images from 800 patients. For multidisease classification, JSIEC (China) and Retina (India) were used. JSIEC includes 1,000 images with 39 categories of common referable ocular disease and conditions. Retina has labels of normal, glaucoma, cataract and other retina disease. The details of datasets, such as imaging devices, country and label category, are listed in Supplementary Table 4. We adhered to all relevant guidelines and ethics regulations in our use of data.

Image generation process

We trained a latent diffusion model (LDM) on real CFPs to generate retinal images. Diffusion model draws inspiration from the diffusion processes in physics and the partial differential equations in mathematics, simulating the random diffusion and denoising of data over time or space. LDM is a specific configuration of the diffusion model in highly compact spaces, enabling high-resolution image generation. Specifically, given a high-dimensional data, LDM first encodes it into a lower-dimensional latent space to get a compact representation $\mathbf{x}$ using a variational autoencoder (VAE⁶¹). The training process⁶² consists of a forward process $\mathbf{q}$ that slowly removes structure at timestep t by adding Gaussian noise to a noisier latent at timestep $t + 1$ and a reverse process modelling by generative model $\mathbf{p}$ that slowly adds structure starting from noisy latent $\mathbf{z}_t$. The LDM is optimized by predicting the added noise content of the latent $\mathbf{z}_t$ during the forward process with a U -Net $\epsilon_\theta(\mathbf{z}_t; t)$, formulated as:

$$L_{\text{diff}}(\theta, \mathbf{x}) = E_{t \sim U(0,1), \epsilon \sim N(\mathbf{0}, \mathbf{I})} \left[w(t) \|\epsilon - \epsilon_\theta(\mathbf{z}_t; t)\|^2 \right] \quad (1)$$

where E is the expectation operator, representing integration over all possible values of t and ϵ , $U(0, 1)$ denotes a uniform distribution between 0 and 1, N represents a standard normal distribution, θ represents the model parameters, $w(t)$ is a weighting function that depends on the timestep t .

We used the stable diffusion model, the latest technique in LDM as our backbone to build a retinal-image diffusion model (Supplementary Fig. 8). The stable diffusion model supports text-to-image generation by using CLIP³³ to introduce a text embedding into the diffusion model as a condition. To incorporate this, we created a corresponding text tag for each of the 150k real retinal images. For images with explicit labels such as glaucoma, the text was set to 'A colour fundus photography retinal image of glaucoma'. For images with no label, the text was set to 'A colour fundus photography retinal image'. We fine-tuned Stable Diffusion v.1.4 on 150k retinal text-to-image pairs. The learning rate was 1×10^{-5} and the batch size was 4. The fine-tuning iteration was 60,000 and it took 2 days on one NVIDIA Tesla A100 (80 GB) GPU. After fine-tuning, we used the retinal-image diffusion model to generate retinal images with a resolution using text-to-image generation. The steps are as follows:

- (1) Text embedding. We controlled the generation of specific categories of retinal images through a text-to-image approach. The text conditions were crafted from the disease category labels of the retinal images. In the data generation process, we first converted the input text prompt $\text{str}_{\text{prompt}}$ into tokens using a tokenizer, and then adopted the pretrained CLIP text encoder to extract text embedding $\mathbf{y}$ with a fixed dimension (77×768 in Stable Diffusion v.1.4). In addition, a blank prompt $\text{str}_{\text{blank}}$ was introduced as an unconditional embedding $\mathbf{y}_{\text{uncond}}$ to generate a latent without specific condition.
- (2) Sampling latent vector. In the inference phase, the input latent for U -Net should be a noise vector from a standard Gaussian distribution. Therefore, we directly sampled a vector $\mathbf{z}$ from the standard Gaussian distribution as input, with dimensions of $B \times c \times h \times w$. Here, B is the batch size, c is the number of channels, and h and w represent the height and width, respectively. In Stable Diffusion v.1.4, $c = 4$, $h = w = 64$. To simultaneously

generate conditioned and unconditioned results, we duplicated the latent and input them both into U -Net.

- (3) Diffusion process. Before performing the diffusion process, it is necessary to specify the diffusion steps T and text guidance scale s_{text} . The diffusion process aims to gradually transform the initial random noise latent $\mathbf{z}_0$ into the target image's latent $\mathbf{z}_T$ through T iterative steps. For each step t , we sent the latent from $t - 1$ step and text embeddings into U -Net to estimate the denoised latent. Specifically, we obtained a text-condition latent prediction $\mathbf{z}_t^{\text{text}}$ and an unconditional latent prediction $\mathbf{z}_t^{\text{uncond}}$ on the basis of the input text embedding $\mathbf{y}$ and unconditional embedding $\mathbf{y}_{\text{uncond}}$, respectively. The final latent variable $\mathbf{z}_t$ in the t step was obtained by weighted summation of two latents with text guidance scale:

$$\mathbf{z}_t = \mathbf{z}_t^{\text{uncond}} + s_{\text{text}} (\mathbf{z}_t^{\text{text}} - \mathbf{z}_t^{\text{uncond}}) \quad (2)$$

This process was repeated for T steps to produce the final latent $\mathbf{z}_T$.

- (4) Image generation. The final latent $\mathbf{z}_T$ was passed through the VAE decoder to generate a retinal image with dimensions of $B \times C \times H \times W$. In Stable Diffusion v.1.4, $C = 4$, $H = W = 64$.

Model architecture and implementation

We used the same model architecture and training process as in RETFound for a fair comparison. The model architecture is a specific configuration of the masked autoencoder based on vision transformer⁴⁴, which consists of an encoder and a decoder (Supplementary Fig. 9). The encoder is implemented by a large vision transformer (ViT-large) with 24 transformer blocks, while the decoder is implemented by a small vision transformer (ViT-small) with 8 transformer blocks. During the training, the input image was randomly masked with a mask ratio of 0.75, and then the unmasked patches (patch size 16×16) were sent to the encoder to extract a high-dimensional feature vector of size 1,024. The decoder takes the feature vector as input to reconstruct the complete image, including the masked patches. The training loss of the model was computed only on the masked pixels, comparing the predicted pixel values with the corresponding real pixel values in the original image, using the mean square error as loss function. The pretrained weights on ImageNet-1k were used to initialize model parameters. Unlike RETFound that is pretrained for 800 epochs on 900k real retinal images, RETFound-DE was first pretrained for 600 epochs on -1 million generated retinal images and then pretrained for another 200 epochs on 150k real retinal images. The batch size was 1,792 (8 GPUs $\times$ 224 per GPU) and the first 20 epochs were used for learning rate warming up (from 0 to a learning rate of 1×10^{-4}). All images were resized to 256×256 with cubic interpolation. We followed the same data augmentation as that in the masked autoencoder in model training, including random crop and resizing the cropped patches to 224×224 , random horizontal flipping and image normalization.

Adaptation to downstream tasks

We also used the same strategy to adapt RETFound-DE to downstream tasks. The decoder was discarded and the input image was sent to the encoder to extract high-level features, followed by a multilayer perceptron to predict the probability of disease categories. The final classification result was determined by the category with the highest probability. A label smoothing cross-entropy loss was calculated between the predicted label and actual label to regulate the output distribution. The batch size was 32 and training epoch was 50. Several data augmentation techniques were involved in the fine-tuning process, including image resize, random flipping (horizontal/vertical), greyscale, rotation, colour jitter and image normalization. The model weights with the highest AUROC on the validation set were saved as the model checkpoints for evaluation.

Explanations for fine-tuned models

We used the GradCam⁶³ in our experiments to generate heat maps of fine-tuned RETFound-DE for a certain ocular disease. It works by calculating the gradients of the predicted class with respect to the feature maps in the network and uses these gradients to identify the image regions that are most important for the model's decision. The feature map from the normalization layer of the last transformer block in the ViT-large encoder was chosen to generate the heat map.

Computational resources

We used 8 NVIDIA Tesla A100 (80 GB) for pretraining on the generated and real retinal images. It took RETFound-DE 5 days to train on 1 million generated images and 1 day to train on 150k real retinal images. For fine-tuning RETFound to downstream tasks, we used an NVIDIA GTX 3090 (24 GB). Fine-tuning RETFound-DE for 50 epochs on a dataset of 1,000 images took ~1 h. All the pretraining and fine-tuning processes in our experiments had the same computational cost.

Evaluation and statistical analysis

Task performance evaluation in this study relied on well-established classification metrics, including the AUROC and the AUPR, accuracy, sensitivity and specificity. The AUROC measures the trade-off between the true positive rate (also known as sensitivity) and the false positive rate at various probability thresholds, while AUPR measures the trade-off between precision and recall. Accuracy represents the proportion of correctly predicted samples by the classifier. High sensitivity indicates the model's effectiveness in detecting true positive cases and high specificity indicates the model's capability to effectively rule out healthy samples. For multiclass classification, we calculated the AUROC and AUPR for each disease category and then averaged them to get the general AUROC and AUPR. To ensure robustness and reliability, we trained the model with five different random seeds and calculated the mean and standard deviation of the performance to calculate the standard error using (standard deviation / $\sqrt{5}$). In addition, we established a 95% CI using $1.96 \times$ standard error. We employed two-sided *t*-tests to assess the statistical significance of the performance differences between RETFound-DE and RETFound. We also employed permutation tests to assess the statistical results, as shown in Supplementary Table 9. A *P* value below 0.05 indicates a significant difference, while a *P* value above 0.05 suggests comparable performance.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The main data supporting the results in this study are available within the paper and its Supplementary Information. Data for pretraining can be accessed through the following weblinks: AIROGS (<https://airogs.grand-challenge.org/data-and-challenge>), Kaggle EyePACS (<https://www.kaggle.com/c/diabetic-retinopathy-detection>), DDR (<https://github.com/nkicli/DDR-dataset>), ODIR-2019 (<https://odir2019.grand-challenge.org>). Data for fine-tuning can be accessed through the following weblinks: IDRID (<https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid>), MESSIDOR-2 (<https://www.adcis.net/en/third-party/messidor2>), APTOS-2019 (<https://www.kaggle.com/competitions/aptos2019-blindness-detection/data>), PAPILA (<https://figshare.com/articles/dataset/PAPILA/14798004/1>), Glaucoma Fundus (<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/1YRRAC>), ORIGA (<https://www.kaggle.com/datasets/arnavjain1/glaucoma-datasets>), AREDS (https://www.ncbi.nlm.nih.gov/projects/gap/cgi-bin/study.cgi?study_id=phs000001.v3.p1), JSIEC (<https://zenodo.org/record/3477553>), Retina (<https://www.kaggle.com/datasets/jr2ngb/>

cataractdataset), REFUGE (<https://ieee-dataport.org/documents/refuge-retinal-fundus-glaucoma-challenge>), RIM-ONE-DL (<https://github.com/miag-ull/rim-one-dl?tab=readme-ov-file>), CheXpert (<https://stanfordmlgroup.github.io/competitions/chexpert/>), Shenzhen Hospital CXR Set (<https://data.lhncbc.nlm.nih.gov/public/Tuberculosis-Chest-X-ray-Datasets/Shenzhen-Hospital-CXR-Set/index.html>), TB Chest X-ray database (<https://www.kaggle.com/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset>).

Code availability

The code of RETFound-DE is available at <https://github.com/Jonllysun/DERETFound> (ref. 64), and an online interactive platform is available at <http://fdudml.cn:12001>. We used stable diffusion implemented by diffusers (<https://github.com/huggingface/diffusers>) for the backbone and image generation. The heat maps were generated with GradCam (<https://github.com/jacobgil/pytorch-grad-cam>) and the *t*-SNE visualization was generated with tsne-pytorch (<https://github.com/mxli1990/tsne-pytorch>).

References

1. Zhou, Y. et al. A foundation model for generalizable disease detection from retinal images. *Nature* **622**, 156–163 (2023).
2. Theodoris, C. V. et al. Transfer learning enables predictions in network biology. *Nature* **618**, 616–624 (2023).
3. Huang, Z. et al. A visual-language foundation model for pathology image analysis using medical Twitter. *Nat. Med.* **29**, 2307–2316 (2023).
4. Zhang, X. et al. Knowledge-enhanced visual-language pre-training on chest radiology images. *Nat. Commun.* **14**, 4542 (2023).
5. Moor, M. et al. Foundation models for generalist medical artificial intelligence. *Nature* **616**, 259–265 (2023).
6. Krishnan, R., Rajpurkar, P. & Topol, E. J. Self-supervised learning in medicine and healthcare. *Nat. Biomed. Eng.* **6**, 1346–1352 (2022).
7. Mitchell, M., Jain, R. & Langer, R. Engineering and physical sciences in oncology: challenges and opportunities. *Nat. Rev. Cancer* **17**, 659–675 (2017).
8. Villoslada, P., Baeza-Yates, R. & Masdeu, J. C. Reclassifying neurodegenerative diseases. *Nat. Biomed. Eng.* **4**, 759–760 (2020).
9. Rajpurkar, P. et al. AI in health and medicine. *Nat. Med.* **28**, 31–38 (2022).
10. Ribaric, S., Ariyaeinia, A. & Pavesic, N. De-identification for privacy protection in multimedia content: a survey. *Signal Process. Image Commun.* **47**, 131–151 (2016).
11. Chang, Q. et al. Mining multi-center heterogeneous medical data with distributed synthetic learning. *Nat. Commun.* **14**, 5510 (2023).
12. Bond-Taylor, S., Leach, A., Long, Y. & Willcocks, C. G. Deep generative modelling: a comparative review of vaes, gans, normalizing flows, energy-based and autoregressive models. *IEEE Trans. Pattern Anal. Mach. Intell.* **44**, 7327–7347 (2021).
13. Kingma, D. P. & Welling, M. Auto-encoding variational Bayes. Preprint at <https://arxiv.org/abs/1312.6114> (2013).
14. Goodfellow, I. et al. Generative adversarial networks. *Commun. ACM* **63**, 139–144 (2020).
15. Ho, J., Jain, A. & Abbeel, P. Denoising diffusion probabilistic models. *Adv. Neural Inf. Process. Syst.* **33**, 6840–6851 (2020).
16. Rombach, R., Blattmann, A., Lorenz, D., Esser, P. & Ommer, B. High-resolution image synthesis with latent diffusion models. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 10684–10695 (IEEE, 2022).
17. Kazerouni, A. et al. Diffusion models in medical imaging: a comprehensive survey. *Med. Image Anal.* **88**, 102846 (2023).
18. Repecka, D. et al. Expanding functional protein sequence spaces using generative adversarial networks. *Nat. Mach. Intell.* **3**, 324–333 (2021).
19. Shin, J. E. et al. Protein design and variant prediction using autoregressive generative models. *Nat. Commun.* **12**, 2403 (2021).

20. Watson, J. L. et al. De novo design of protein structure and function with RFdiffusion. *Nature* **620**, 1089–1100 (2023).
21. Schmitt, L. T. et al. Prediction of designer-recombinases for DNA editing with generative deep learning. *Nat. Commun.* **13**, 7966 (2022).
22. Godinez, W. J. et al. Design of potent antimalarials with generative chemistry. *Nat. Mach. Intell.* **4**, 180–186 (2022).
23. Huang, X. et al. The landscape of mRNA nanomedicine. *Nat. Med.* **28**, 2273–2287 (2022).
24. Chen, Z. et al. A deep generative model for molecule optimization via one fragment modification. *Nat. Mach. Intell.* **3**, 1040–1049 (2021).
25. Zhong, W., Yang, Z. & Chen, C. Y. C. Retrosynthesis prediction using an end-to-end graph generative architecture for molecular graph editing. *Nat. Commun.* **14**, 3009 (2023).
26. Das, P. et al. Accelerated antimicrobial discovery via deep generative models and molecular dynamics simulations. *Nat. Biomed. Eng.* **5**, 613–623 (2021).
27. Kanakasabapathy, M. K. et al. Adaptive adversarial neural networks for the analysis of lossy and domain-shifted datasets of medical images. *Nat. Biomed. Eng.* **5**, 571–585 (2021).
28. Ozyoruk, K. B. et al. A deep-learning model for transforming the style of tissue images from cryosectioned to formalin-fixed and paraffin-embedded. *Nat. Biomed. Eng.* **6**, 1407–1419 (2022).
29. DeGrave, A. J. et al. Auditing the inference processes of medical-image classifiers by leveraging generative AI and the expertise of physicians. *Nat. Biomed. Eng.* <https://doi.org/10.1038/s41551-023-01160-9> (2023).
30. Cao, R. et al. Label-free intraoperative histology of bone tissue via deep-learning-assisted ultraviolet photoacoustic microscopy. *Nat. Biomed. Eng.* **7**, 124–134 (2023).
31. Nichol, A. et al. Glide: towards photorealistic image generation and editing with text-guided diffusion models. Preprint at <https://arxiv.org/abs/2112.10741> (2021).
32. Ramesh, A. et al. Zero-shot text-to-image generation. In *Proc. International Conference on Machine Learning* 8821–8831 (PMLR, 2021).
33. Radford, A. et al. Learning transferable visual models from natural language supervision. In *Proc. International Conference on Machine Learning* 8748–8763 (PMLR, 2021).
34. Kather, J. N. et al. Medical domain knowledge in domain-agnostic generative AI. *npj Digit. Med.* **5**, 90 (2022).
35. Burlina, P. M. et al. Assessment of deep generative models for high-resolution synthetic retinal image generation of age-related macular degeneration. *JAMA Ophthalmol.* **137**, 258–264 (2019).
36. Yoon, J. et al. EHR-Safe: generating high-fidelity and privacy-preserving synthetic electronic health records. *npj Digit. Med.* **6**, 141 (2023).
37. Trabucco, B., Doherty, K., Gurinas, M. & Salakhutdinov, R. Effective data augmentation with diffusion models. In *Proc. International Conference on Learning Representations (ICLR, 2024)*.
38. Zhang, A. et al. Shifting machine learning for healthcare from development to deployment and from models to data. *Nat. Biomed. Eng.* **6**, 1330–1345 (2022).
39. Chen, R. J. et al. Synthetic data in machine learning for medicine and healthcare. *Nat. Biomed. Eng.* **5**, 493–497 (2021).
40. DuMont Schütte, A. et al. Overcoming barriers to data sharing with medical image generation: a comprehensive evaluation. *npj Digit. Med.* **4**, 141 (2021).
41. *World Report on Vision* (World Health Organization, 2019).
42. Cen, L. P. et al. Automatic detection of 39 fundus diseases and conditions in retinal photographs using deep neural networks. *Nat. Commun.* **12**, 4828 (2021).
43. Alimanov, A. & Islam, M. B. Denoising diffusion probabilistic model for retinal image generation and segmentation. In *Proc. IEEE International Conference on Computational Photography* 1–12 (IEEE, 2023).
44. Dosovitskiy, A. et al. An image is worth 16x16 words: transformers for image recognition at scale. In *Proc. International Conference on Learning Representations (ICLR, 2021)*.
45. He, K. et al. Masked autoencoders are scalable vision learners. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 16000–16009 (IEEE, 2022).
46. Karthik, M. & Sohler, D. APTOS 2019 Blindness Detection (Kaggle, 2019).
47. Porwal, P. et al. Idris: diabetic retinopathy–segmentation and grading challenge. *Med. Image Anal.* **59**, 101561 (2020).
48. Decencière, E. et al. Feedback on a publicly distributed image database: the Messidor database. *Image Anal. Stereol.* **33**, 231–234 (2014).
49. Kovalyk, O. et al. PAPILA: dataset with fundus images and clinical data of both eyes of the same patient for glaucoma assessment. *Sci. Data* **9**, 291 (2022).
50. Zhang, Z. et al. Origa-light: an online retinal fundus image database for glaucoma analysis and research. In *Proc. Annual International Conference of the IEEE Engineering in Medicine and Biology* 3065–3068 (IEEE, 2010).
51. Irvin, J. et al. Chexpert: a large chest radiograph dataset with uncertainty labels and expert comparison. In *Proc. AAAI Conference on Artificial Intelligence* Vol. 33 590–597 (AAAI, 2019).
52. Jaeger, S. et al. Two public chest X-ray datasets for computer-aided screening of pulmonary diseases. *Quant. Imaging Med. Surg.* **4**, 475–477 (2014).
53. Rahman, T. et al. Reliable tuberculosis detection using chest X-ray with deep learning, segmentation and visualization. *IEEE Access* **8**, 191586–191601 (2020).
54. Peng, W., Adeli, E., Zhao, Q. & Pohl, K. M. in *Medical Image Computing and Computer Assisted Intervention* 14–24 (MICCAI, 2023).
55. Eschweiler, D. et al. Denoising diffusion probabilistic models for generation of realistic fully-annotated microscopy image datasets. *PLoS Comput. Biol.* **20**, e1011890 (2024).
56. Ktena, I. et al. Generative models improve fairness of medical classifiers under distribution shifts. *Nat. Med.* **30**, 1166–1173 (2024).
57. Bachmann, R., Mizrahi, D., Atanov, A. & Zamir, A. Multimae: multi-modal multi-task masked autoencoders. In *Proc. European Conference on Computer Vision* 348–367 (Springer, 2022).
58. Shumailov, I. et al. AI models collapse when trained on recursively generated data. *Nature* **631**, 755–759 (2024).
59. Yang, Y. et al. The limits of fair medical imaging AI in real-world generalization. *Nat. Med.* **30**, 2838–2848 (2024).
60. de Vente, C. et al. AIROGS: artificial intelligence for robust glaucoma screening challenge. *IEEE Trans. Med. Imaging* **43**, 542–557 (2024).
61. van den Oord, A., Vinyals, O. & Kavukcuoglu, K. Neural discrete representation learning. In *Proc. 31st Conference on Neural Information Processing Systems* 6309–6318 (NIPS, 2017).
62. Song, Y. et al. Score-based generative modeling through stochastic differential equations. In *Proc. International Conference on Learning Representations (ICLR, 2021)*.
63. Selvaraju, R. R. et al. Grad-cam: visual explanations from deep networks via gradient-based localization. In *Proc. IEEE International Conference on Computer Vision* 618–626 (IEEE, 2017).
64. Yuqi. Controllable generative model enables high data efficiency for building medical foundation model. *GitHub* <https://github.com/Jonlysun/DERETFound> (2024).

Acknowledgements

We acknowledge support for this work provided by the National Natural Science Foundation of China (grant numbers U2001209, 62372117 and 62472102) and the Natural Science Foundation of Shanghai

(grant number 21ZR1406600). The computations in this research were performed using the CFFF platform of Fudan University. The AREDS dataset used for the analyses described in this paper was obtained from the Age-Related Eye Disease Study (AREDS) Database found at https://www.ncbi.nlm.nih.gov/projects/gap/cgi-bin/study.cgi?study_id=phs000001.v3.p1 through dbGaP accession number phs000001.v3.p1. Funding support for AREDS was provided by the National Eye Institute (N01-EY-0-2127). We thank the AREDS participants and the AREDS Research Group for their valuable contribution to this research.

Author contributions

B.Y. and W.T. supervised the research. B.Y. conceived the technique. Y.S. implemented the algorithm. Y.S. and W.T. designed the validation experiments. Y.S. trained the network and performed the validation experiments. Y.S. and W.T. analysed the validation results. Z.G. verified the code. R.H. provided technical support on the implementation of the web page. Z.G., M.P. and S.C. collected the public datasets. Y.S., W.T. and B.Y. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41551-025-01365-0>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41551-025-01365-0>.

Correspondence and requests for materials should be addressed to Bo Yan.

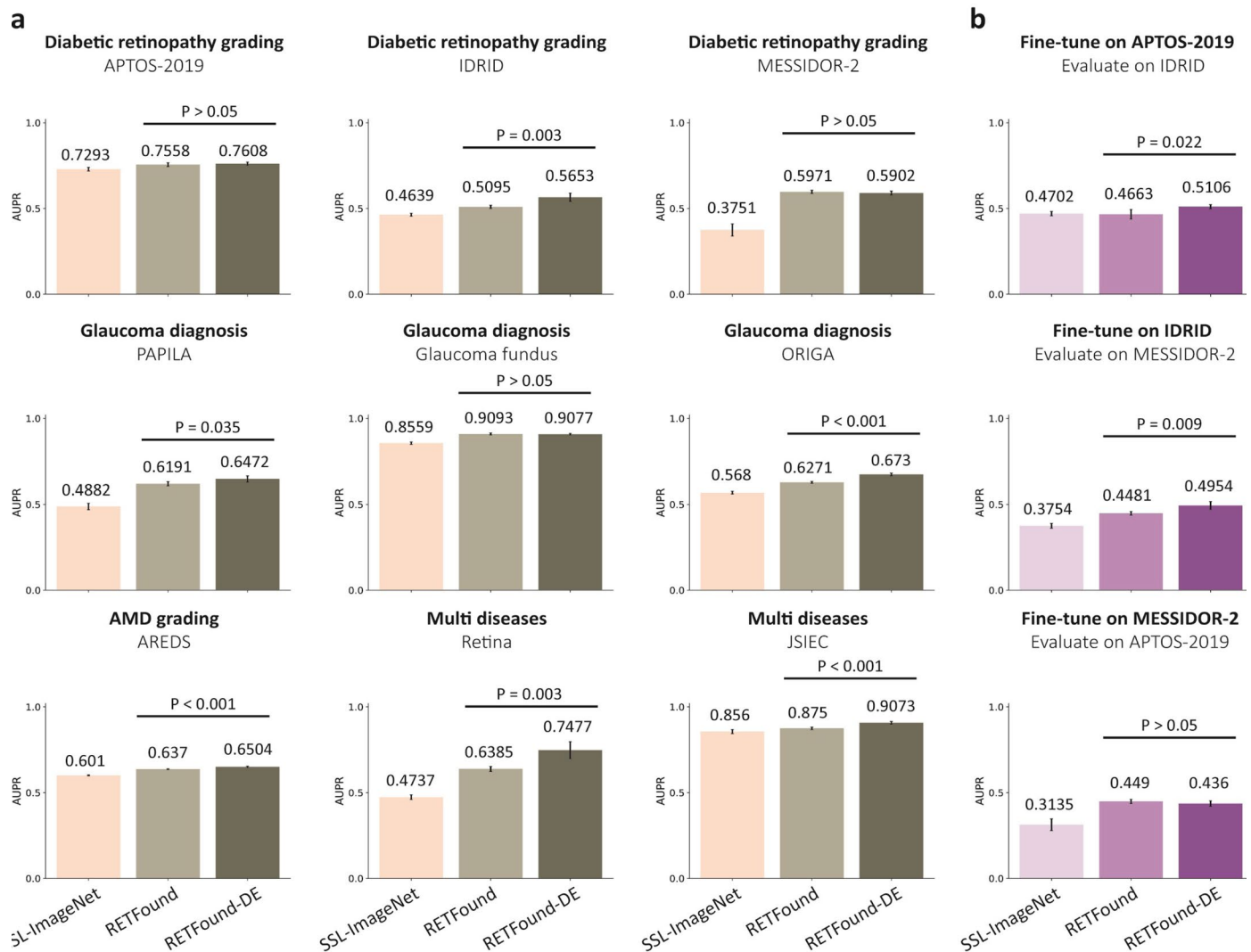
Peer review information *Nature Biomedical Engineering* thanks Pearse Keane and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

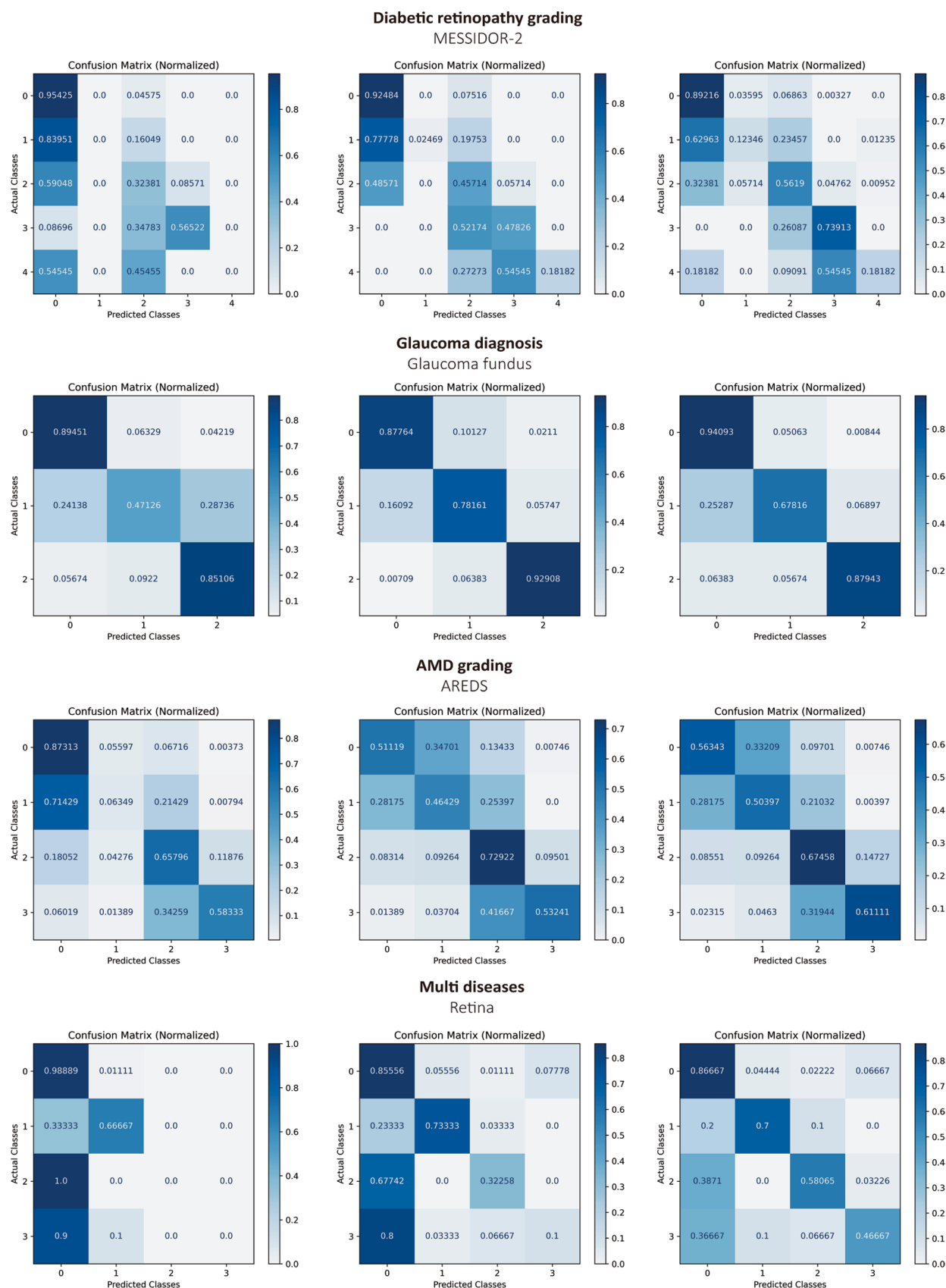
Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2025



Extended Data Fig. 1 | Performance (AUPR) in downstream tasks. a, Internal evaluation. We fine-tuned the pretrained models on nine public datasets across four downstream tasks: diabetic retinopathy grading, glaucoma diagnosis, age-related macular degeneration (AMD) grading and multi diseases classification. Compared to RETFound, RETFound-DE achieves superior performance on six datasets ($P < 0.05$) and comparable performance on the other three datasets

($P > 0.05$). **b**, External evaluation. Models are fine-tuned on one diabetic retinopathy grading dataset and evaluated on the others. RETFound-DE outperforms RETFound when fine-tuned on APTOS-2019 and evaluated on IDRID, or when fine-tuned on IDRID and evaluated on MESSIDOR-2. We present the mean value of AUPR on each bar and the error bars show 95% confidence intervals. P-value was calculated with the two-sided t-test and listed in the figure.



SSL-ImageNet

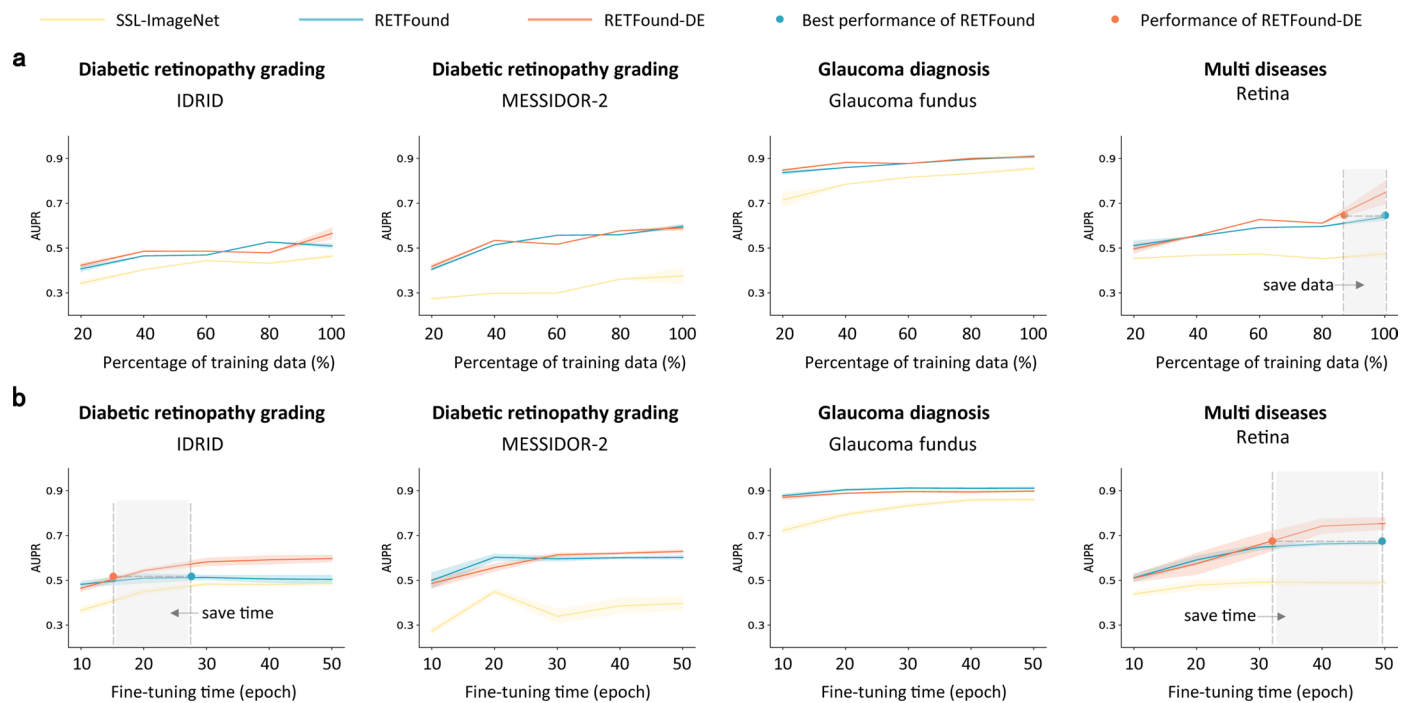
RETFound

RETFound-DE

Extended Data Fig. 2 | Confusion matrices comparison in downstream tasks.

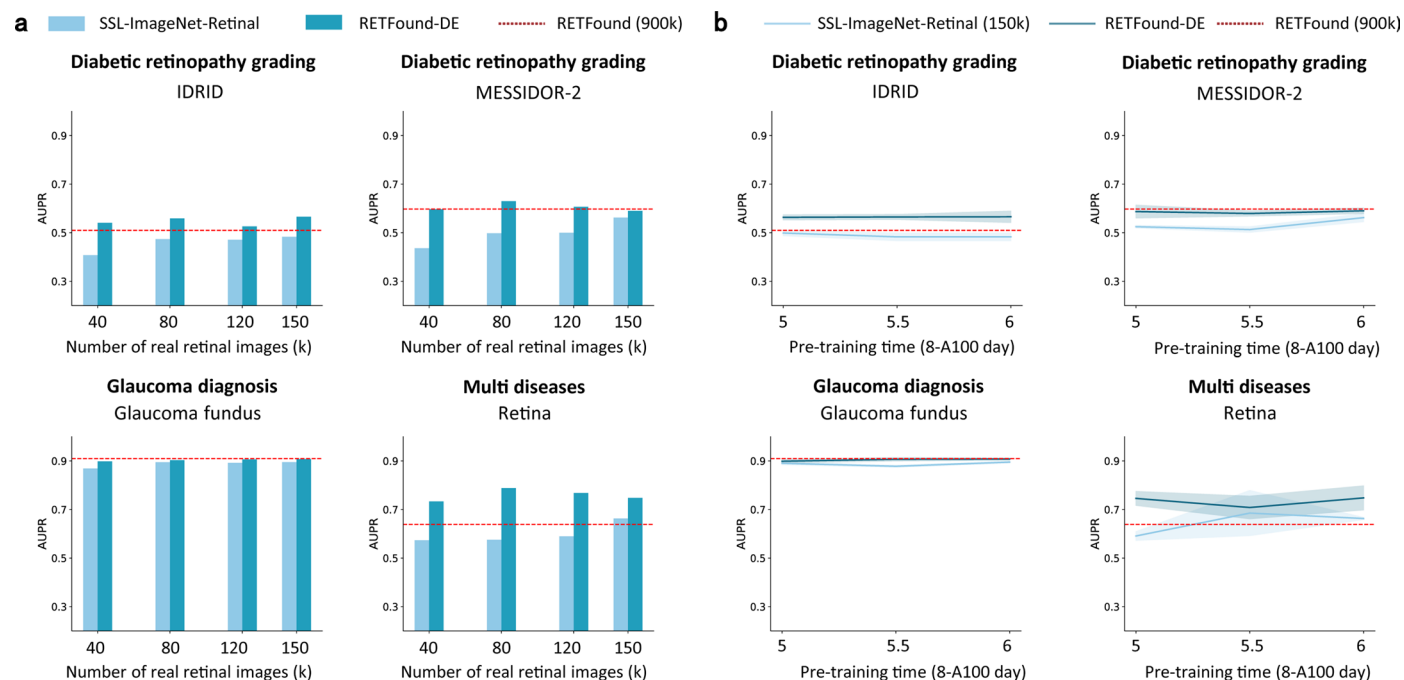
The confusion matrices show the comparison between the predicted classes and the actual labels by the models, with each element of the matrix representing the prediction distribution for a specific class. RETFound-DE's diagonal prediction

scores are generally higher than SSL-ImageNet, and even higher than RETFound, indicating that it classifies each class more accurately in every task and has a lower misclassification rate.



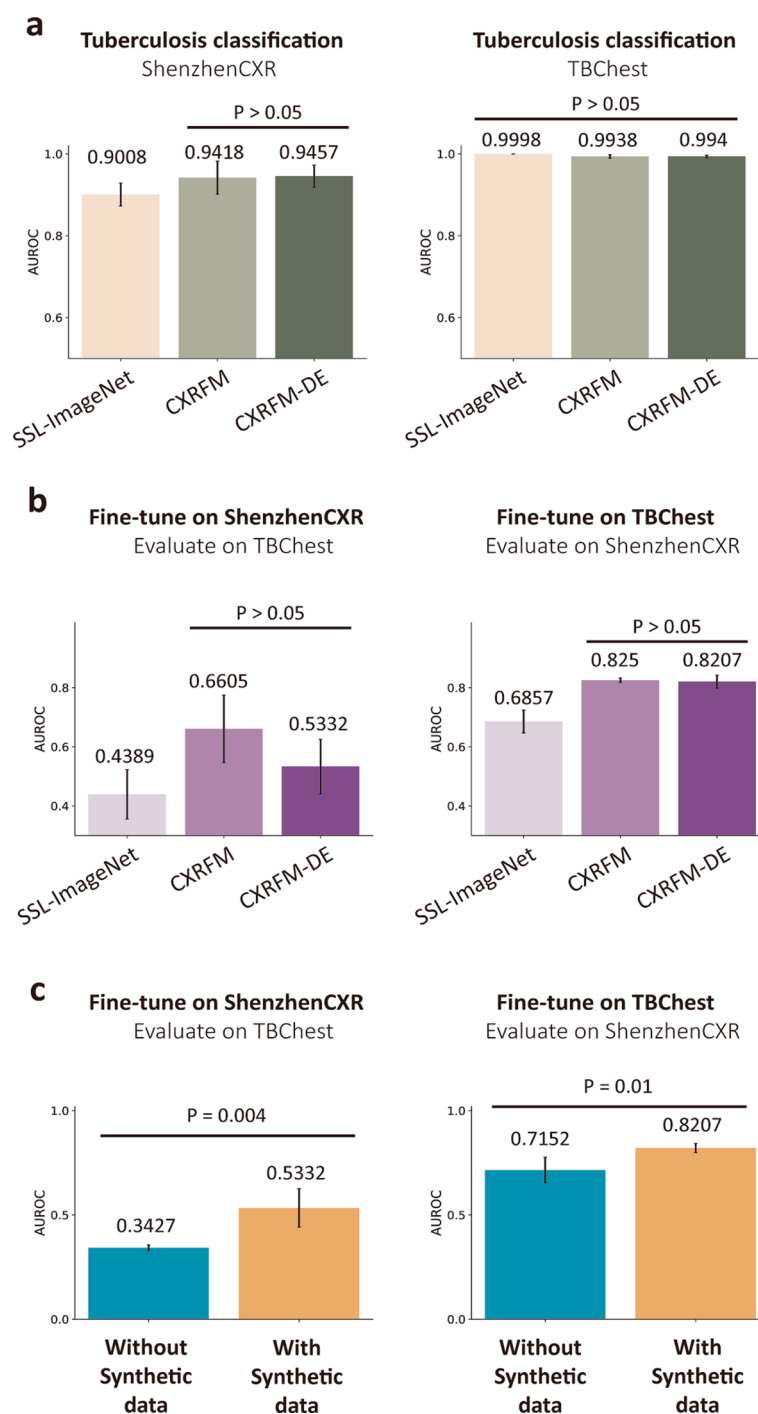
Extended Data Fig. 3 | Label and training time efficiency (AUPR). **a**, Label efficiency. Label efficiency refers to the amount of training data and labels to achieve the target performance for a deep learning networks. RETFound-DE and RETFound show superior label efficiency than SSL-ImageNet. RETFound-DE even outperforms RETFound on Retina dataset. **b**, Training time efficiency. Training

time efficiency refers to the fine-tuning time to achieve the target performance for foundation models when adapted to downstream datasets. RETFound-DE enables same performance as RETFound in less training time on several datasets, such as IDRID and Retina. The dashed grey lines highlight the significant difference between RETFound and RETFound-DE.



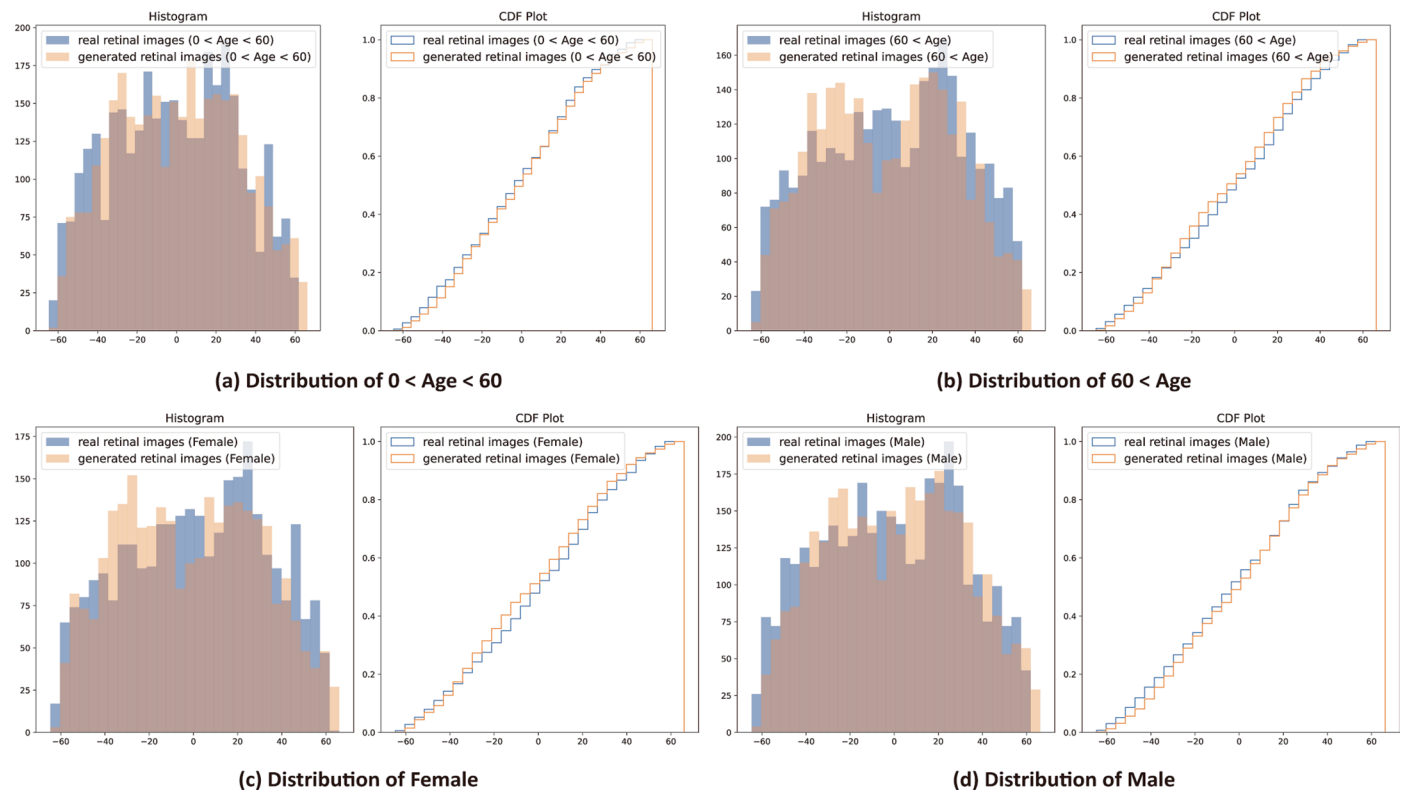
Extended Data Fig. 4 | Performance and pretraining time comparisons of SSL-ImageNet-Retinal and RETFound-DE (AUPR). **a**, The effect of synthetic data on using different number of real data for pretraining. The efficacy of SSL-ImageNet-Retinal progressively enhances on four downstream tasks as the quantity of real retinal images used for pretraining increases. By pretraining on generated images, RETFound-DE shows a significant performance improvement than SSL-ImageNet-Retinal. On IDRID, MESSIDOR-2 and Retina datasets,

RETFound-DE outperforms RETFound when pretrained on only 40k real retinal images. **b**, The performance of RETFound-DE and SSL-ImageNet-Retinal (150k) over a matched computational time from 5 to 6 8-A100 days. We use 8-A100 day as a unit to denote the pretraining time of using 8 NVIDIA A100 GPUs for one day. For both models, the pretraining dataset at this time period is 150k real retinal image dataset. RETFound-DE consistently outperforms SSL-ImageNet-Retinal (150k) on all four downstream datasets within the same pretraining time.



Extended Data Fig. 5 | Internal and external evaluation on Chest X-ray images. **a**, Internal evaluation. CXRFM and CXRFM-DE perform comparably on the ShenzhenCXR dataset and significantly outperformed SSL-ImageNet. On the TBChest dataset, all three models achieved similar performance, with an AUROC exceeding 0.99. **b**, External evaluation. When fine-tuned on TBChest and evaluated on ShenzhenCXR, CXRFM and CXRFM-DE perform similarly, both

substantially surpassing SSL-ImageNet. When fine-tuned on ShenzhenCXR and evaluated on TBChest, CXRFM outperforms CXRFM-DE. **c**, CXRFM-DE (denoted as 'With synthetic data') significantly outperforms SSL-ImageNet-Chest (20k) (denoted as 'Without synthetic data') under both conditions, demonstrating the enhancement in generalization brought about by synthetic data.



Extended Data Fig. 6 | Feature distribution of real and synthetic datasets in terms of age and gender. We use histograms and cumulative distribution functions to illustrate the feature distributions of real data and synthetic data. (a) and (b) show that the features distribution of real and synthetic data for the

$0 < \text{Age} < 60$ and $\text{Age} > 60$ groups. (c) and (d) show that the features distribution of real and synthetic data for female and male. The results show the consistency of feature distribution between real and synthetic datasets in terms of age and gender.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	The code used to train, fine-tune and evaluate RETFound-DE is available at https://github.com/Jonllysun/DERETFound , and is based on Pytorch.
Data analysis	Data were analysed with Python v3.8 (https://www.python.org/), SciPy v1.5.4 (https://www.scipy.org/), pandas v2.0.3 (https://github.com/pandas-dev/pandas), Scikit-Learn v0.24.2 (https://scikit-learn.org/stable), Pillow v8.3.2 (https://pypi.org/project/Pillow). Heatmaps were generated with GradCAM (https://github.com/hila-chefer/Transformer-Explainability). The stable diffusion model was implemented via diffusers (https://github.com/huggingface/diffusers). The t-SNE visualization was generated with tsne-pytorch (https://github.com/mxl1990/tsne-pytorch).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The main data supporting the results in this study are available within the paper and Supplementary Information. Data for pretraining can be accessed through the following weblinks: AIROGS (<https://airogs.grand-challenge.org/data-and-challenge>), Kaggle EyePACS (<https://www.kaggle.com/c/diabetic-retinopathy-detection>), DDR (<https://github.com/nkicsl/DDR-dataset>), ODIR-2019 (<https://odir2019.grand-challenge.org>). Data for fine-tuning can be accessed through the following weblinks: IDRID (<https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid>), MESSIDOR-2 (<https://www.adcis.net/en/third-party/messidor2>), APTOS-2019 (<https://www.kaggle.com/competitions/aptos2019-blindness-detection/data>), PAPILA (<https://figshare.com/articles/dataset/PAPILA/14798004/1>), Glaucoma Fundus (<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/1YRRAC>), ORIGA (<https://www.kaggle.com/datasets/arnavjain1/glaucoma-datasets>), AREDs (https://www.ncbi.nlm.nih.gov/projects/gap/cgi-bin/study.cgi?study_id=phs000001.v3.p1), JSIEC (<https://zenodo.org/record/3477553>), Retina (<https://www.kaggle.com/datasets/jr2ngb/cataractdataset>), REFUGE (<https://ieee-dataport.org/documents/refuge-retinal-fundus-glaucoma-challenge>), RIM-ONE-DL (<https://github.com/miag-ull/rim-one-dl?tab=readme-ov-file>), CheXpert (<https://stanfordmlgroup.github.io/competitions/chexpert/>), Shenzhen Hospital CXR Set (<https://data.lhncbc.nlm.nih.gov/public/Tuberculosis-Chest-X-ray-Datasets/Shenzhen-Hospital-CXR-Set/index.html>), TB Chest X-ray database (<https://www.kaggle.com/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset>).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

We collected 150,786 CFPs from multiple publicly available datasets. Of these datasets, 101,442 images were from AIROGS, a large dataset with 54,274 patients for glaucoma screening. The age distribution of AIROGS has a mean value of 56.7 and a standard deviation of 10.2.

Reporting on race, ethnicity, or other socially relevant groupings

Not applicable.

Population characteristics

We collected 150,786 CFPs from multiple publicly available datasets, including AIROGS\cite{devente23airogs}, Kaggle EyePACS, DDR and ODIR-2019. Of these datasets, 101,442 images were from AIROGS, a large dataset with 54,274 patients for glaucoma screening. 35,126 and 7,218 images were from Kaggle EyePACS and DDR for diabetic retinopathy grading. 7,000 images were from ODIR-2019, a multi-disease classification dataset including cataract, myopia, diabetic retinopathy, AMD and glaucoma. Images in AIROGS and Kaggle EyePACS datasets are from the United States of America, whereas DDR and ODIR-2019 are from China.

Recruitment

Not applicable.

Ethics oversight

Not applicable.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

We generated 1.2 million retinal images for pretraining foundation models. The real retinal images for pretraining are 150,786 CFPs.

Data exclusions

Generated colour fundus photography images with poor quality.

Replication

The codes of DERETFound are freely available at <https://github.com/Jonllysun/DERETFound>. The results were confirmed 5 times with 5 different random seeds.

Randomization

The training/validation/testing data for downstream tasks were randomly splitted according to these ratios: 55%:15%:30%.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging